Laser Cell 2

Bevel-capable 2D laser cutting machine (6 kW fiber)
Operating and Maintenance Manual

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Prepared by production engineering for internal use.

Synthetic document written for a student project. It is not a TRUMPF manual or any manufacturer's manual. Machine name, codes, part numbers and values are illustrative.

Revision history

Rev	Date	Change
1	May 2023	First issue.
2	Feb 2024	Nozzle: replace every 80-100 operating hours. Protective glass: check once per week.
3	Mar 2026	Nozzle interval and protective glass check changed after quality review, see section 8.1. Rev 2 values are withdrawn. Appendix C added for the Cell 1 machine.

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1 Safety

This chapter applies to everyone who operates, cleans, maintains or repairs the machine. Read it before you work on the machine for the first time. The machine contains a class 4 laser inside a class 1 enclosure.

1.1 Laser radiation

Record any repair of the chiller in the maintenance book. The chiller is monitored by the control; faults are shown on the operator console. The position of the chiller is shown in the overview drawing in chapter 2.

Spare parts for the slat table are listed in Appendix B. Messages from the slat table are listed in chapter 9 under their code. The slat table is cleaned with a dry cloth; do not use solvents.

DANGER	Invisible laser radiation. Never look into the beam or its reflections. Do not bypass the door interlock.

1. Verify that the guard covers are closed and locked.
2. Inspect that the display shows no open messages.
3. Look at that the guard covers are closed and locked.
4. Look at that the guard covers are closed and locked.
5. Look at that the sensor surface is clean and dry.
6. Make sure that the fastening screws are in place.
7. Make sure that the drive belt has the correct tension.

1.2 Fumes and dust

A label on the compressed air unit shows the part number and the year of manufacture. Record any repair of the compressed air unit in the maintenance book. The compressed air unit was upgraded in 2024; older drawings may show the previous version.

The cable carrier has its own fuse in the control cabinet. Wear gloves when you touch the cable carrier; edges can be sharp. If the cable carrier does not respond, switch the machine off at the main switch and inform the supervisor.

WARNING	Cutting fumes are harmful. Operate only with the fume extraction unit running.

1. Confirm that the guard covers are closed and locked.
2. Look at that the fastening screws are in place.
3. Look at that the fastening screws are in place.
4. Inspect that the status lamp is green.
5. Check that the display shows no open messages.
6. Confirm that the emergency stop is released.
7. Check that no foreign objects lie in the travel range.

1.3 Fire

The central lubrication unit can be moved by hand only when the drives are switched off. The central lubrication unit is monitored by the control; faults are shown on the operator console. Keep the cover of the central lubrication unit closed during automatic operation.

The service technician checks the cable carrier during the yearly inspection. The cable carrier reaches its operating condition a few minutes after the machine is switched on. After a repair of the cable carrier, run the machine for one test program without a sheet.

WARNING	Sheet residues and oily scrap can ignite. Empty the scrap conveyor every shift.

1. Look at that the reference marks line up.
2. Inspect that the reference marks line up.
3. Verify that the display shows no open messages.
4. Check that no foreign objects lie in the travel range.
5. Inspect that the status lamp is green.
6. Confirm that the display shows no open messages.
7. Inspect that the fastening screws are in place.

1.4 Electrical energy

Changes to the central lubrication unit must be approved by production engineering. The operating hours of the central lubrication unit are shown on the Diagnostics screen. If the central lubrication unit does not respond, switch the machine off at the main switch and inform the supervisor.

Before working on the chiller, switch off the machine and secure the main switch with your own padlock. Dust and spatter shorten the service life of the chiller; keep the area clean. If the chiller does not respond, switch the machine off at the main switch and inform the supervisor.

DANGER	Lethal voltage in the control cabinet. Only electricians may open it.

1. Make sure that the reference marks line up.
2. Make sure that the filter indicator is in the green range.
3. Check that the reference marks line up.
4. Inspect that the drive belt has the correct tension.

1.5 Moving axes

Dust and spatter shorten the service life of the cable carrier; keep the area clean. Spare parts for the cable carrier are listed in Appendix B. Record any repair of the cable carrier in the maintenance book.

Keep the cover of the pallet changer closed during automatic operation. The pallet changer can be moved by hand only when the drives are switched off. The pallet changer is designed for continuous operation in a three-shift system.

WARNING	Axes move without warning in automatic mode. Stay outside the light curtain.

1. Verify that the drive belt has the correct tension.
2. Confirm that the sensor surface is clean and dry.
3. Look at that the fastening screws are in place.
4. Inspect that the cable connections are tight.
5. Check that the display shows no open messages.

1.6 Compressed gases

Spare parts for the pallet changer are listed in Appendix B. A label on the pallet changer shows the part number and the year of manufacture. The pallet changer is designed for continuous operation in a three-shift system.

Before working on the operator console, switch off the machine and secure the main switch with your own padlock. The operator console is cleaned with a dry cloth; do not use solvents. Spare parts for the operator console are listed in Appendix B.

WARNING	Nitrogen displaces oxygen. Ventilate the room and secure gas cylinders.

1. Make sure that the display shows no open messages.
2. Make sure that the emergency stop is released.
3. Confirm that the reference marks line up.
4. Make sure that the filter indicator is in the green range.

1.7 Hot parts

If the control cabinet makes unusual noise, stop the machine and inform the supervisor. The control cabinet was upgraded in 2024; older drawings may show the previous version. The control cabinet can be moved by hand only when the drives are switched off.

Settings of the beam guide housing are stored in the machine data and should not be changed by the operator. The beam guide housing is cleaned with a dry cloth; do not use solvents. Messages from the beam guide housing are listed in chapter 9 under their code.

NOTE	Freshly cut parts can be hot. Wear gloves when removing parts.

1. Look at that no foreign objects lie in the travel range.
2. Confirm that the sensor surface is clean and dry.
3. Check that the status lamp is green.
4. Look at that the filter indicator is in the green range.

1.8 Noise

The chiller is monitored by the control; faults are shown on the operator console. Messages from the chiller are listed in chapter 9 under their code. Wear gloves when you touch the chiller; edges can be sharp.

Settings of the chiller are stored in the machine data and should not be changed by the operator. The chiller can be moved by hand only when the drives are switched off. The chiller is designed for continuous operation in a three-shift system.

NOTE	Wear hearing protection during long nitrogen cuts with high pressure.

1. Check that the fastening screws are in place.
2. Make sure that the emergency stop is released.
3. Confirm that the hoses show no kinks or abrasion.
4. Look at that the cable connections are tight.
5. Make sure that the status lamp is green.

1.9 Personal protective equipment

Task	Safety shoes	Gloves	Glasses	Hearing protection
Loading sheets	yes	cut-resistant	no	no
Removing parts	yes	cut-resistant	yes	no
Changing nozzle	yes	yes	yes	long N2 jobs
Cleaning optics	yes	cut-resistant	yes	no
Emptying scrap	yes	yes	no	yes

Task	Safety shoes	Gloves	Glasses	Hearing protection
Gas cylinder change	yes	cut-resistant	no	yes
Lubrication	yes	yes	yes	yes
Cleaning the slat table	yes	cut-resistant	no	yes

2 Machine description

2.1 Overview

Laser Cell 2 is a flatbed laser cutting machine with a 6 kW fiber laser, an automatic pallet changer and a tilting cutting head for bevel cuts up to 45 degrees. It cuts mild steel with oxygen (O2) and stainless steel and aluminium with nitrogen (N2).

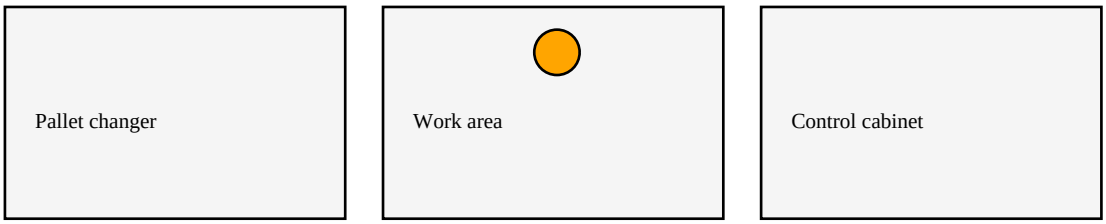


Figure 2.1 Machine overview (top view, not to scale).

2.2 Technical data

Item	Value
Working range X / Y / Z	3000 / 1500 / 115 mm
Max. sheet weight	900 kg
Laser power	6000 W (fiber)
Max. bevel angle	+/- 45 deg
Simultaneous axis speed	140 m/min
Positioning accuracy	+/- 0.05 mm
Max. N2 inlet pressure	30 bar
Max. O2 inlet pressure	8 bar
Power supply	400 V, 3 phase, 50 Hz
Mass	approx. 11 t

2.3 Pallet changer

Dust and spatter shorten the service life of the pallet changer; keep the area clean. The pallet changer has its own fuse in the control cabinet. A label on the pallet changer shows the part number and the year of manufacture. If the pallet changer makes unusual noise, stop the machine and inform the supervisor. Before working on the pallet changer, switch off the machine and secure the main switch with your own padlock.

The pallet changer was upgraded in 2024; older drawings may show the previous version. Changes to the pallet changer must be approved by production engineering. Record any repair of the pallet changer in the maintenance book. Messages from the pallet changer are listed in chapter 9 under their code.

Parameter	Value
Pallet changer parameter 1	417 mm
Pallet changer parameter 2	861 kg
Pallet changer parameter 3	143 bar
Pallet changer parameter 4	270 l/min

2.4 Fume extraction unit

After a repair of the fume extraction unit, run the machine for one test program without a sheet. The fume extraction unit is cleaned with a dry cloth; do not use solvents. Before working on the fume extraction unit, switch off the machine and secure the main switch with your own padlock. The fume extraction unit was upgraded in 2024; older drawings may show the previous version. If the fume extraction unit does not respond, switch the machine off at the main switch and inform the supervisor.

The operating hours of the fume extraction unit are shown on the Diagnostics screen. The fume extraction unit is designed for continuous operation in a three-shift system. The position of the fume extraction unit is shown in the overview drawing in chapter 2. Wear gloves when you touch the fume extraction unit; edges can be sharp.

Parameter	Value
Fume extraction unit parameter 1	119 s
Fume extraction unit parameter 2	279 l/min
Fume extraction unit parameter 3	348 l/min
Fume extraction unit parameter 4	404 bar
Fume extraction unit parameter 5	500 bar
Fume extraction unit parameter 6	735 l/min

2.5 Chiller

Messages from the chiller are listed in chapter 9 under their code. Spare parts for the chiller are listed in Appendix B. The chiller can be moved by hand only when the drives are switched off. Only trained personnel may open the chiller. Keep the cover of the chiller closed during automatic operation.

Changes to the chiller must be approved by production engineering. Wear gloves when you touch the chiller; edges can be sharp. The operating hours of the chiller are shown on the Diagnostics screen. The chiller can be moved by hand only when the drives are switched off.

Parameter	Value
Chiller parameter 1	471 bar
Chiller parameter 2	339 bar
Chiller parameter 3	512 mm
Chiller parameter 4	16 l/min
Chiller parameter 5	421 s
Chiller parameter 6	324 l/min

2.6 Operator console

After a repair of the operator console, run the machine for one test program without a sheet. The operator console can be moved by hand only when the drives are switched off. Keep the cover of the operator console closed during automatic operation. The position of the operator console is shown in the overview drawing in chapter 2. Do not use compressed air to clean the operator console; it blows dust into the bearings.

The position of the operator console is shown in the overview drawing in chapter 2. Record any repair of the operator console in the maintenance book. Messages from the operator console are listed in chapter 9 under their code. The operator console is cleaned with a dry cloth; do not use solvents.

Parameter	Value
Operator console parameter 1	313 mm
Operator console parameter 2	522 kg

Parameter	Value
Operator console parameter 3	214 bar
Operator console parameter 4	807 s
Operator console parameter 5	256 mm
Operator console parameter 6	235 kg

2.7 Light curtain

Before working on the light curtain, switch off the machine and secure the main switch with your own padlock. The light curtain reaches its operating condition a few minutes after the machine is switched on. The position of the light curtain is shown in the overview drawing in chapter 2. Do not use compressed air to clean the light curtain; it blows dust into the bearings. The light curtain is designed for continuous operation in a three-shift system.

The light curtain is cleaned with a dry cloth; do not use solvents. The light curtain is monitored by the control; faults are shown on the operator console. Spare parts for the light curtain are listed in Appendix B. The light curtain has its own fuse in the control cabinet.

Parameter	Value
Light curtain parameter 1	274 mm
Light curtain parameter 2	40 W
Light curtain parameter 3	544 W

2.8 Scrap conveyor

After a repair of the scrap conveyor, run the machine for one test program without a sheet. Changes to the scrap conveyor must be approved by production engineering. The scrap conveyor reaches its operating condition a few minutes after the machine is switched on. The operating hours of the scrap conveyor are shown on the Diagnostics screen. Spare parts for the scrap conveyor are listed in Appendix B.

Spare parts for the scrap conveyor are listed in Appendix B. Settings of the scrap conveyor are stored in the machine data and should not be changed by the operator. The scrap conveyor is designed for continuous operation in a three-shift system. The scrap conveyor reaches its operating condition a few minutes after the machine is switched on.

Parameter	Value
Scrap conveyor parameter 1	808 kg
Scrap conveyor parameter 2	828 bar
Scrap conveyor parameter 3	150 bar
Scrap conveyor parameter 4	766 mm
Scrap conveyor parameter 5	262 kg
Scrap conveyor parameter 6	823 s

2.9 Central lubrication unit

The central lubrication unit is designed for continuous operation in a three-shift system. If the central lubrication unit makes unusual noise, stop the machine and inform the supervisor. Record any repair of the central lubrication unit in the maintenance book. Only trained personnel may open the central lubrication unit. The operating hours of the central lubrication unit are shown on the Diagnostics screen.

Record any repair of the central lubrication unit in the maintenance book. Changes to the central lubrication unit must be approved by production engineering. Wear gloves when you touch the central lubrication unit; edges can be sharp. Do not use compressed air to clean the central lubrication unit; it blows dust into the

bearings.

Parameter	Value
Central lubrication unit parameter 1	540 bar
Central lubrication unit parameter 2	56 kg
Central lubrication unit parameter 3	256 kg
Central lubrication unit parameter 4	856 l/min
Central lubrication unit parameter 5	48 mm

3 Transport and installation

3.1 Lifting and transport

The Z axis drive is monitored by the control; faults are shown on the operator console. If the Z axis drive makes unusual noise, stop the machine and inform the supervisor. The Z axis drive can be moved by hand only when the drives are switched off. Only trained personnel may open the Z axis drive.

If the bevel rotation axis makes unusual noise, stop the machine and inform the supervisor. If the bevel rotation axis does not respond, switch the machine off at the main switch and inform the supervisor. The bevel rotation axis can be moved by hand only when the drives are switched off. Only trained personnel may open the bevel rotation axis.

1. Look at that the cable connections are tight.
2. Check that the emergency stop is released.
3. Make sure that the fastening screws are in place.
4. Make sure that the hoses show no kinks or abrasion.
5. Look at that the guard covers are closed and locked.
6. Verify that the display shows no open messages.
7. Check that the display shows no open messages.

3.2 Floor and foundation

The chiller is designed for continuous operation in a three-shift system. The chiller has its own fuse in the control cabinet. Changes to the chiller must be approved by production engineering. Messages from the chiller are listed in chapter 9 under their code.

Wear gloves when you touch the central lubrication unit; edges can be sharp. Only trained personnel may open the central lubrication unit. The central lubrication unit is designed for continuous operation in a three-shift system. Spare parts for the central lubrication unit are listed in Appendix B.

1. Confirm that the emergency stop is released.
2. Make sure that the sensor surface is clean and dry.
3. Check that the display shows no open messages.
4. Verify that the cable connections are tight.
5. Look at that the filter indicator is in the green range.
6. Verify that the filter indicator is in the green range.
7. Confirm that the sensor surface is clean and dry.
8. Confirm that the status lamp is green.

3.3 Electrical connection

Settings of the chiller are stored in the machine data and should not be changed by the operator. The chiller is designed for continuous operation in a three-shift system. Only trained personnel may open the chiller. If the chiller makes unusual noise, stop the machine and inform the supervisor.

Settings of the part removal flap are stored in the machine data and should not be changed by the operator. The part removal flap is monitored by the control; faults are shown on the operator console. The part removal flap is cleaned with a dry cloth; do not use solvents. The part removal flap reaches its operating condition a few minutes after the machine is switched on.

1. Inspect that the reference marks line up.
2. Make sure that the display shows no open messages.
3. Check that the filter indicator is in the green range.

4. Check that the emergency stop is released.

5. Look at that the filter indicator is in the green range.

3.4 Gas connection

Changes to the part removal flap must be approved by production engineering. Dust and spatter shorten the service life of the part removal flap; keep the area clean. If the part removal flap makes unusual noise, stop the machine and inform the supervisor. The part removal flap is cleaned with a dry cloth; do not use solvents.

Messages from the network interface are listed in chapter 9 under their code. Only trained personnel may open the network interface. Changes to the network interface must be approved by production engineering. The network interface has its own fuse in the control cabinet.

1. Look at that the emergency stop is released.

2. Inspect that the status lamp is green.

3. Verify that the drive belt has the correct tension.

4. Inspect that the hoses show no kinks or abrasion.

5. Check that the fastening screws are in place.

6. Check that the cable connections are tight.

7. Inspect that the drive belt has the correct tension.

8. Verify that the guard covers are closed and locked.

9. Make sure that the hoses show no kinks or abrasion.

3.5 Network connection

The sheet clamp unit can be moved by hand only when the drives are switched off. The sheet clamp unit reaches its operating condition a few minutes after the machine is switched on. The sheet clamp unit is monitored by the control; faults are shown on the operator console. The position of the sheet clamp unit is shown in the overview drawing in chapter 2.

Changes to the Z axis drive must be approved by production engineering. Before working on the Z axis drive, switch off the machine and secure the main switch with your own padlock. Only trained personnel may open the Z axis drive. The Z axis drive has its own fuse in the control cabinet.

1. Make sure that the fastening screws are in place.

2. Make sure that the guard covers are closed and locked.

3. Confirm that the sensor surface is clean and dry.

4. Verify that the reference marks line up.

5. Inspect that the hoses show no kinks or abrasion.

3.6 First start-up by service

A label on the X axis drive shows the part number and the year of manufacture. After a repair of the X axis drive, run the machine for one test program without a sheet. Settings of the X axis drive are stored in the machine data and should not be changed by the operator. The X axis drive is monitored by the control; faults are shown on the operator console.

Record any repair of the chiller in the maintenance book. Spare parts for the chiller are listed in Appendix B. The chiller can be moved by hand only when the drives are switched off. The chiller is monitored by the control; faults are shown on the operator console.

1. Make sure that the guard covers are closed and locked.

2. Inspect that the drive belt has the correct tension.

3. Verify that the hoses show no kinks or abrasion.

4. Confirm that the filter indicator is in the green range.
5. Make sure that the hoses show no kinks or abrasion.
6. Look at that the hoses show no kinks or abrasion.
7. Inspect that the display shows no open messages.
8. Verify that the guard covers are closed and locked.
9. Make sure that the display shows no open messages.

4 Operator console and software

The operator console has a touch screen with six main screens. Buttons are listed per screen below.

4.1 Home screen

The Home screen is opened from the main menu. Do not use compressed air to clean the operator console; it blows dust into the bearings. The operator console is monitored by the control; faults are shown on the operator console. The operator console can be moved by hand only when the drives are switched off.

Button	Function
HOME-01	Resets the sheet data
HOME-02	Filters the pallet status
HOME-03	Starts the job list
HOME-04	Starts the shift report
HOME-05	Filters the shift report
HOME-06	Starts the counter values
HOME-07	Resets the sheet data
HOME-08	Exports the user list
HOME-09	Exports the sheet data
HOME-10	Filters the sheet data
HOME-11	Filters the counter values
HOME-12	Shows the message list
HOME-13	Exports the counter values

4.2 Production screen

The Production screen is opened from the main menu. Only trained personnel may open the operator console. The operator console is monitored by the control; faults are shown on the operator console. The operator console can be moved by hand only when the drives are switched off.

Button	Function
PROD-01	Starts the pallet status
PROD-02	Filters the job list
PROD-03	Resets the job list
PROD-04	Exports the pallet status
PROD-05	Resets the job list
PROD-06	Starts the user list
PROD-07	Starts the message list
PROD-08	Shows the shift report

4.3 Programs screen

The Programs screen is opened from the main menu. Wear gloves when you touch the operator console; edges can be sharp. If the operator console makes unusual noise, stop the machine and inform the supervisor. Messages from the operator console are listed in chapter 9 under their code.

Button	Function
PROG-01	Opens the counter values
PROG-02	Exports the job list
PROG-03	Resets the pallet status
PROG-04	Resets the message list
PROG-05	Filters the user list

Button	Function
PROG-06	Opens the shift report
PROG-07	Starts the shift report
PROG-08	Opens the axis positions
PROG-09	Starts the message list
PROG-10	Shows the sheet data
PROG-11	Shows the job list
PROG-12	Exports the counter values
PROG-13	Exports the user list

4.4 Technology tables screen

The Technology tables screen is opened from the main menu. Messages from the operator console are listed in chapter 9 under their code. Do not use compressed air to clean the operator console; it blows dust into the bearings. Only trained personnel may open the operator console.

Button	Function
TECH-01	Exports the counter values
TECH-02	Filters the job list
TECH-03	Opens the user list
TECH-04	Filters the sheet data
TECH-05	Starts the message list
TECH-06	Shows the user list
TECH-07	Opens the pallet status
TECH-08	Opens the sheet data
TECH-09	Resets the shift report
TECH-10	Exports the message list
TECH-11	Filters the shift report
TECH-12	Shows the job list

4.5 Diagnostics screen

The Diagnostics screen is opened from the main menu. The service technician checks the operator console during the yearly inspection. Changes to the operator console must be approved by production engineering. The operating hours of the operator console are shown on the Diagnostics screen.

Button	Function
DIAG-01	Opens the message list
DIAG-02	Resets the message list
DIAG-03	Exports the counter values
DIAG-04	Filters the sheet data
DIAG-05	Shows the job list
DIAG-06	Filters the shift report
DIAG-07	Exports the message list
DIAG-08	Shows the user list
DIAG-09	Opens the counter values
DIAG-10	Shows the sheet data
DIAG-11	Starts the axis positions

4.6 Settings screen

The Settings screen is opened from the main menu. Do not use compressed air to clean the operator console; it blows dust into the bearings. The operator console is cleaned with a dry cloth; do not use solvents. Record any repair of the operator console in the maintenance book.

Button	Function
SETT-01	Resets the counter values
SETT-02	Shows the message list
SETT-03	Starts the axis positions
SETT-04	Starts the counter values
SETT-05	Resets the shift report
SETT-06	Resets the job list
SETT-07	Exports the shift report
SETT-08	Opens the shift report

4.7 User management screen

The User management screen is opened from the main menu. Only trained personnel may open the operator console. The service technician checks the operator console during the yearly inspection. Settings of the operator console are stored in the machine data and should not be changed by the operator.

Button	Function
USER-01	Shows the axis positions
USER-02	Opens the axis positions
USER-03	Starts the counter values
USER-04	Opens the axis positions
USER-05	Shows the job list
USER-06	Filters the shift report
USER-07	Filters the axis positions
USER-08	Resets the sheet data
USER-09	Starts the message list
USER-10	Filters the counter values
USER-11	Shows the message list
USER-12	Shows the axis positions
USER-13	Opens the sheet data

4.8 Pallet management screen

The Pallet management screen is opened from the main menu. If the operator console makes unusual noise, stop the machine and inform the supervisor. Before working on the operator console, switch off the machine and secure the main switch with your own padlock. The operator console is monitored by the control; faults are shown on the operator console.

Button	Function
PALL-01	Opens the counter values
PALL-02	Shows the sheet data
PALL-03	Filters the user list
PALL-04	Filters the pallet status
PALL-05	Starts the sheet data
PALL-06	Shows the message list
PALL-07	Starts the pallet status
PALL-08	Starts the user list
PALL-09	Resets the sheet data

Button	Function
PALL-10	Shows the job list
PALL-11	Resets the pallet status

5 Operation

5.1 Switching on

The position of the chiller is shown in the overview drawing in chapter 2. Settings of the chiller are stored in the machine data and should not be changed by the operator. Do not use compressed air to clean the chiller; it blows dust into the bearings. The chiller is cleaned with a dry cloth; do not use solvents.

1. Inspect that the reference marks line up.
2. Inspect that the emergency stop is released.
3. Verify that the hoses show no kinks or abrasion.
4. Verify that the display shows no open messages.
5. Look at that the display shows no open messages.

5.2 Loading a sheet

A label on the Z axis drive shows the part number and the year of manufacture. The service technician checks the Z axis drive during the yearly inspection. If the Z axis drive makes unusual noise, stop the machine and inform the supervisor. The position of the Z axis drive is shown in the overview drawing in chapter 2.

1. Look at that no foreign objects lie in the travel range.
2. Inspect that no foreign objects lie in the travel range.
3. Make sure that the sensor surface is clean and dry.
4. Confirm that the sensor surface is clean and dry.
5. Inspect that the fastening screws are in place.
6. Verify that no foreign objects lie in the travel range.
7. Check that the sensor surface is clean and dry.
8. Confirm that the display shows no open messages.

5.3 Selecting a program

Keep the cover of the scrap conveyor closed during automatic operation. The scrap conveyor is cleaned with a dry cloth; do not use solvents. Before working on the scrap conveyor, switch off the machine and secure the main switch with your own padlock. The scrap conveyor has its own fuse in the control cabinet.

1. Make sure that the cable connections are tight.
2. Verify that the guard covers are closed and locked.
3. Look at that the filter indicator is in the green range.
4. Verify that the display shows no open messages.
5. Make sure that no foreign objects lie in the travel range.
6. Look at that the emergency stop is released.
7. Confirm that the guard covers are closed and locked.

5.4 Feed rate and released programs

Use the cutting tables in chapter 6 as the starting point for every job. Small changes are allowed to compensate for material batches. Do not increase feed rate above the cutting table value by more than 10 % without a test cut.

Program suffix	Meaning	Released
(none)	Standard technology	yes
_TEST	Test cut, single part	yes
_BEV	Bevel technology	yes
_FAST*	Increased feed	no

* Programs ending in _FAST are not released by engineering. Report any _FAST program found on the machine.

5.5 Pallet change

The service technician checks the bevel rotation axis during the yearly inspection. Record any repair of the bevel rotation axis in the maintenance book. Spare parts for the bevel rotation axis are listed in Appendix B. The bevel rotation axis is monitored by the control; faults are shown on the operator console.

A label on the sheet clamp unit shows the part number and the year of manufacture. After a repair of the sheet clamp unit, run the machine for one test program without a sheet. The sheet clamp unit is monitored by the control; faults are shown on the operator console.

1. Inspect that the cable connections are tight.

2. Look at that the hoses show no kinks or abrasion.

3. Look at that the reference marks line up.

4. Confirm that the filter indicator is in the green range.

5. Check that the guard covers are closed and locked.

6. Confirm that the fastening screws are in place.

7. Verify that the emergency stop is released.

5.6 Removing parts

Before working on the part removal flap, switch off the machine and secure the main switch with your own padlock. Messages from the part removal flap are listed in chapter 9 under their code. The part removal flap reaches its operating condition a few minutes after the machine is switched on. The part removal flap has its own fuse in the control cabinet.

Dust and spatter shorten the service life of the bevel rotation axis; keep the area clean. The operating hours of the bevel rotation axis are shown on the Diagnostics screen. The position of the bevel rotation axis is shown in the overview drawing in chapter 2.

1. Verify that the emergency stop is released.

2. Look at that the display shows no open messages.

3. Check that the sensor surface is clean and dry.

4. Confirm that no foreign objects lie in the travel range.

5.7 Ending the shift

The fume extraction unit was upgraded in 2024; older drawings may show the previous version. If the fume extraction unit does not respond, switch the machine off at the main switch and inform the supervisor. After a repair of the fume extraction unit, run the machine for one test program without a sheet. The fume extraction unit is cleaned with a dry cloth; do not use solvents.

Do not use compressed air to clean the control cabinet; it blows dust into the bearings. Changes to the control cabinet must be approved by production engineering. Spare parts for the control cabinet are listed in Appendix B.

1. Inspect that the reference marks line up.

2. Make sure that no foreign objects lie in the travel range.

3. Make sure that the filter indicator is in the green range.
4. Look at that the display shows no open messages.

6 Cutting technology

6.1 Cutting gases

Oxygen (O2) burns the material and adds energy; it is used for mild steel. Nitrogen (N2) only blows the melt out and gives an oxide-free edge; it is used for stainless steel and aluminium and needs high pressure.

6.2 Focus position and nozzles

A negative focus value lies below the sheet surface. Nitrogen cuts use larger nozzles than oxygen cuts. The nozzle diameter for each thickness is given in the cutting tables.

6.3 Cutting table: Mild steel S235JR (O2)

Starting values for Mild steel S235JR (O2). Thicknesses not listed must be cut with a test program first.

Thickness (mm)	Power (W)	Feed (mm/min)	Pressure (bar)	Focus (mm)	Nozzle (mm)
3	3200	3600	0.8	3.0	1.2
6	4000	2600	0.7	3.5	1.2
10	5000	1700	0.6	4.0	1.4

Do not use compressed air to clean the beam guide housing; it blows dust into the bearings. Settings of the beam guide housing are stored in the machine data and should not be changed by the operator. The beam guide housing reaches its operating condition a few minutes after the machine is switched on.

Table continued:

Thickness (mm)	Power (W)	Feed (mm/min)	Pressure (bar)	Focus (mm)	Nozzle (mm)
12	6000	1500	0.6	4.5	1.4
15	6000	1100	0.5	5.0	1.7

6.4 Cutting table: Stainless steel 1.4301 (N2)

Starting values for Stainless steel 1.4301 (N2). Thicknesses not listed must be cut with a test program first.

Thickness (mm)	Power (W)	Feed (mm/min)	Pressure (bar)	Focus (mm)	Nozzle (mm)
2	4000	9000	14.0	-1.0	2.0
4	6000	5200	16.0	-2.0	2.3
6	6000	3200	18.0	-3.5	2.3

Changes to the beam guide housing must be approved by production engineering. The beam guide housing was upgraded in 2024; older drawings may show the previous version. Keep the cover of the beam guide housing closed during automatic operation.

Table continued:

Thickness (mm)	Power (W)	Feed (mm/min)	Pressure (bar)	Focus (mm)	Nozzle (mm)
8	6000	2000	19.0	-5.0	2.7
10	6000	1300	20.0	-6.5	3.0

6.5 Cutting table: Aluminium AlMg3 (N2)

Starting values for Aluminium AlMg3 (N2). Thicknesses not listed must be cut with a test program first.

Thickness (mm)	Power (W)	Feed (mm/min)	Pressure (bar)	Focus (mm)	Nozzle (mm)
3	5000	7000	14.0	-1.5	2.0
5	6000	4200	16.0	-3.0	2.3

The operating hours of the beam guide housing are shown on the Diagnostics screen. Spare parts for the beam guide housing are listed in Appendix B. Record any repair of the beam guide housing in the maintenance book.

Table continued:

Thickness (mm)	Power (W)	Feed (mm/min)	Pressure (bar)	Focus (mm)	Nozzle (mm)
8	6000	1900	18.0	-5.0	2.7

6.6 Pierce parameters

Piercing values are stored with the technology table and are not changed by the operator. Wear gloves when you touch the operator console; edges can be sharp. Before working on the operator console, switch off the machine and secure the main switch with your own padlock. Dust and spatter shorten the service life of the operator console; keep the area clean.

7 Bevel cutting

7.1 Bevel types

Spare parts for the bevel rotation axis are listed in Appendix B. A label on the bevel rotation axis shows the part number and the year of manufacture. Changes to the bevel rotation axis must be approved by production engineering. The bevel rotation axis has its own fuse in the control cabinet.

Messages from the bevel rotation axis are listed in chapter 9 under their code. The bevel rotation axis is monitored by the control; faults are shown on the operator console. Only trained personnel may open the bevel rotation axis.

7.2 Bevel programs

The bevel rotation axis is designed for continuous operation in a three-shift system. Keep the cover of the bevel rotation axis closed during automatic operation. Only trained personnel may open the bevel rotation axis. Do not use compressed air to clean the bevel rotation axis; it blows dust into the bearings.

Keep the cover of the bevel rotation axis closed during automatic operation. The service technician checks the bevel rotation axis during the yearly inspection. The bevel rotation axis can be moved by hand only when the drives are switched off.

7.3 Measuring the first part

A label on the bevel rotation axis shows the part number and the year of manufacture. Dust and spatter shorten the service life of the bevel rotation axis; keep the area clean. After a repair of the bevel rotation axis, run the machine for one test program without a sheet. If the bevel rotation axis makes unusual noise, stop the machine and inform the supervisor.

The bevel rotation axis reaches its operating condition a few minutes after the machine is switched on. The service technician checks the bevel rotation axis during the yearly inspection. The bevel rotation axis is designed for continuous operation in a three-shift system.

7.4 After a collision

A collision of the cutting head (message 3301) stops the program. The collision can shift the bevel axis and the capacitive height sensor out of calibration, so bevel parts cut afterwards will have the wrong angle.

1. Check nozzle, ceramic ring and head for damage.
2. Calibrate the height sensor.
3. Have the service technician calibrate the bevel axis.
4. Cut one bevel test part and measure it with a protractor.

WARNING	After any collision calibrate height sensor and bevel axis before bevel jobs.

8 Maintenance

8.1 Maintenance intervals

The intervals below are valid from Revision 3. They replace the values of Revision 2.

Item	Interval	Note
Protective glass	Check every shift; clean when spatter visible	Replace if burnt spots or cracks
Nozzle	Inspect daily; replace every 40-60 operating hours or when damaged	N2 cutting is most sensitive to nozzle wear. Keep spare 2.3 mm and 2.7 mm nozzles.
Nozzle centring	After every nozzle change	Automatic check (7300)
Height sensor / bevel axis	After any collision; otherwise monthly	Service technician for bevel axis
Cutting gas supply	Weekly check of tank level and regulator	Facilities team

8.2 Nozzle

The nozzle guides the cutting gas onto the kerf. A damaged or worn nozzle gives an uneven gas jet.

1. Move the head to the change position.
2. Unscrew the nozzle by hand.
3. Screw in the new nozzle of the diameter given in the cutting table.
4. Start nozzle centring; the machine shows message 7300 when it has passed.



Figure 8.1 Nozzle seen from below. Left: new. Right: worn, orifice no longer round. A worn nozzle causes dross on nitrogen cuts and rough edges; replace the nozzle.

8.3 Protective glass

The protective glass is a cheap window below the focusing lens. Spatter on the glass absorbs laser power, heats the glass and can crack it. Message 2105 asks for cleaning; message 4001 (laser power reduced) is usually dirty optics as well.

1. Switch the laser to standby.
2. Pull out the protective glass drawer.
3. Hold the glass at the edges and look through it against a light.
4. Clean or replace protective glass: clean with lint-free wipes and optics cleaner; replace it if there are burnt spots or cracks.
5. Push the drawer back and acknowledge the message.

8.4 Cutting gas supply

The facilities team checks the tank level and the regulator every week. A regulator fault shows as messages 1201 and 1202. To fix a low or unstable supply: check tank, regulator and lines.

8.5 Central lubrication unit

The central lubrication unit is designed for continuous operation in a three-shift system. After a repair of the central lubrication unit, run the machine for one test program without a sheet. Before working on the central lubrication unit, switch off the machine and secure the main switch with your own padlock. The position of the central lubrication unit is shown in the overview drawing in chapter 2.

1. Confirm that the cable connections are tight.
2. Inspect that the sensor surface is clean and dry.
3. Check that the display shows no open messages.
4. Check that no foreign objects lie in the travel range.
5. Inspect that the sensor surface is clean and dry.

Message 8000 reminds the operator to lubricate. Lubricate at end of shift.

8.6 Fume extraction unit

The fume extraction unit is monitored by the control; faults are shown on the operator console. Spare parts for the fume extraction unit are listed in Appendix B. The fume extraction unit has its own fuse in the control cabinet. After a repair of the fume extraction unit, run the machine for one test program without a sheet.

1. Make sure that the drive belt has the correct tension.
2. Verify that the filter indicator is in the green range.
3. Make sure that the emergency stop is released.
4. Verify that the guard covers are closed and locked.
5. Verify that the filter indicator is in the green range.
6. Inspect that the fastening screws are in place.
7. Confirm that the emergency stop is released.

8.7 Chiller

Record any repair of the chiller in the maintenance book. The position of the chiller is shown in the overview drawing in chapter 2. The operating hours of the chiller are shown on the Diagnostics screen. Before working on the chiller, switch off the machine and secure the main switch with your own padlock.

1. Check that the emergency stop is released.
2. Confirm that the reference marks line up.
3. Confirm that the filter indicator is in the green range.
4. Make sure that the emergency stop is released.
5. Look at that the fastening screws are in place.
6. Verify that the emergency stop is released.
7. Inspect that the filter indicator is in the green range.

8.8 Slat table

Before working on the slat table, switch off the machine and secure the main switch with your own padlock. Messages from the slat table are listed in chapter 9 under their code. Keep the cover of the slat table closed during automatic operation. Only trained personnel may open the slat table.

1. Look at that the filter indicator is in the green range.
2. Confirm that the fastening screws are in place.
3. Check that the display shows no open messages.
4. Make sure that the hoses show no kinks or abrasion.
5. Look at that the hoses show no kinks or abrasion.
6. Verify that the cable connections are tight.
7. Make sure that the filter indicator is in the green range.
8. Verify that the emergency stop is released.

8.9 Scrap conveyor

The scrap conveyor is designed for continuous operation in a three-shift system. Changes to the scrap conveyor must be approved by production engineering. The scrap conveyor can be moved by hand only when the drives are switched off. Spare parts for the scrap conveyor are listed in Appendix B.

1. Verify that the emergency stop is released.
2. Inspect that the status lamp is green.
3. Verify that the emergency stop is released.
4. Inspect that the guard covers are closed and locked.
5. Verify that the sensor surface is clean and dry.
6. Make sure that the guard covers are closed and locked.
7. Verify that the drive belt has the correct tension.
8. Verify that the display shows no open messages.
9. Look at that the filter indicator is in the green range.

8.10 Compressed air unit

After a repair of the compressed air unit, run the machine for one test program without a sheet. The compressed air unit was upgraded in 2024; older drawings may show the previous version. Changes to the compressed air unit must be approved by production engineering. The operating hours of the compressed air unit are shown on the Diagnostics screen.

1. Look at that the status lamp is green.
2. Verify that the sensor surface is clean and dry.
3. Verify that the emergency stop is released.
4. Verify that the hoses show no kinks or abrasion.
5. Look at that the cable connections are tight.
6. Verify that no foreign objects lie in the travel range.
7. Check that the emergency stop is released.

8.11 Light curtain

Dust and spatter shorten the service life of the light curtain; keep the area clean. The light curtain reaches its operating condition a few minutes after the machine is switched on. A label on the light curtain shows the part number and the year of manufacture. The light curtain can be moved by hand only when the drives are switched off.

1. Look at that the cable connections are tight.
2. Make sure that the hoses show no kinks or abrasion.

3. Make sure that the cable connections are tight.
4. Look at that the sensor surface is clean and dry.
5. Make sure that the cable connections are tight.
6. Check that the reference marks line up.

8.12 Cable carrier

Before working on the cable carrier, switch off the machine and secure the main switch with your own padlock. The operating hours of the cable carrier are shown on the Diagnostics screen. The service technician checks the cable carrier during the yearly inspection. The cable carrier reaches its operating condition a few minutes after the machine is switched on.

1. Confirm that the reference marks line up.
2. Check that the emergency stop is released.
3. Verify that the status lamp is green.
4. Confirm that the reference marks line up.
5. Make sure that the cable connections are tight.
6. Inspect that the display shows no open messages.

9 Machine messages

The machine writes every message to the message log with a timestamp. INFO messages are normal. WARNING messages need action before the next job. ERROR messages stop the program. The list is sorted by code.

Code	Level	Text	Action
1041	ERROR	Gas: parameter missing	Restart the program.
1201	WARNING	Cutting gas pressure below setpoint	Supply pressure too low. Check tank, regulator and lines. Expect dross on N2 cuts.
1202	INFO	Cutting gas pressure fluctuation	Short pressure drop. Repeated: check regulator.
1203	WARNING	Cutting gas pressure above setpoint	Check regulator setting.
1209	ERROR	Gas: temperature changed	Acknowledge and restart.
1265	INFO	Gas: temperature limit reached	Check the door interlock.
1337	INFO	Gas: status limit reached	Call service.
1372	INFO	Gas: signal limit reached	Normal operation.
1411	WARNING	Gas: counter not plausible	Acknowledge and restart.
1488	INFO	Gas: counter out of range	Restart the program.
1607	INFO	Gas: temperature changed	Check the beam guide housing.
1609	WARNING	Gas: parameter not plausible	Call service.
1614	INFO	Gas: value limit reached	Check the scrap conveyor.
1631	WARNING	Gas: value timeout	Acknowledge and restart.
1640	ERROR	Gas: signal limit reached	Call service.
1652	INFO	Gas: signal timeout	Acknowledge and restart.
1715	INFO	Gas: counter limit reached	Call service.
1738	INFO	Gas: parameter limit reached	Call service.
1891	INFO	Gas: status limit reached	Check the fume extraction unit.
2105	WARNING	Protective glass contamination - check and clean	See section 8.3.
2106	ERROR	Protective glass missing	Insert the protective glass drawer.
2173	INFO	Optics: counter out of range	Acknowledge and restart.
2208	ERROR	Optics: temperature missing	Restart the program.
2224	INFO	Optics: signal timeout	Check the cable carrier.
2252	ERROR	Optics: parameter limit reached	Normal operation.
2280	ERROR	Optics: parameter changed	Restart the program.
2300	INFO	Optics: counter missing	Normal operation.
2317	INFO	Optics: signal not plausible	Check the Y axis drive.
2354	ERROR	Optics: parameter not plausible	Acknowledge and restart.
2406	WARNING	Optics: parameter timeout	Check the door interlock.
2508	INFO	Optics: temperature timeout	Normal operation.
2546	INFO	Optics: parameter limit reached	Acknowledge and restart.
2607	INFO	Optics: value limit reached	Call service.
2695	INFO	Optics: status not plausible	Restart the program.
2704	INFO	Optics: value timeout	Restart the program.
2948	INFO	Optics: value missing	Call service.
2966	WARNING	Optics: temperature timeout	Call service.
3018	INFO	Cutting head: value changed	Call service.
3160	INFO	Cutting head: signal limit reached	Check the slat table.
3191	INFO	Cutting head: parameter not plausible	Acknowledge and restart.
3234	INFO	Cutting head: status changed	Acknowledge and restart.
3243	WARNING	Cutting head: signal not plausible	Call service.

Code	Level	Text	Action
3301	ERROR	Collision detected - cutting head	Program stops. See section 7.4 before restart.
3302	WARNING	Capacitive height sensor - calibration required	Calibrate sensor. Bevel cuts will be wrong until done.
3303	INFO	Capacitive height sensor - signal noisy	Clean the nozzle.
3395	INFO	Cutting head: signal out of range	Restart the program.
3405	WARNING	Cutting head: value limit reached	Normal operation.
3539	ERROR	Cutting head: status out of range	Restart the program.
3718	WARNING	Cutting head: parameter changed	Check the pallet changer.
3898	WARNING	Cutting head: signal out of range	Check the part removal flap.
3917	INFO	Cutting head: temperature out of range	Normal operation.
3975	INFO	Cutting head: status not plausible	Acknowledge and restart.
3978	ERROR	Cutting head: status out of range	Acknowledge and restart.
3989	INFO	Cutting head: temperature limit reached	Call service.
4001	WARNING	Laser power reduced - beam path monitoring	Usually dirty optics. See section 8.3; if it stays, call service.
4002	WARNING	Laser power reduced - chiller temperature	Check the chiller.
4106	ERROR	Laser: signal missing	Normal operation.
4128	WARNING	Laser: status not plausible	Check the Y axis drive.
4143	INFO	Laser: counter not plausible	Normal operation.
4191	WARNING	Laser: counter limit reached	Restart the program.
4227	INFO	Laser: status limit reached	Normal operation.
4374	WARNING	Laser: value timeout	Restart the program.
4380	INFO	Laser: temperature missing	Normal operation.
4526	ERROR	Laser: signal limit reached	Acknowledge and restart.
4708	INFO	Laser: temperature out of range	Check the beam guide housing.
4709	WARNING	Laser: parameter out of range	Restart the program.
4726	INFO	Laser: signal changed	Call service.
4738	INFO	Laser: counter timeout	Acknowledge and restart.
4741	INFO	Laser: temperature timeout	Restart the program.
4767	INFO	Laser: counter changed	Acknowledge and restart.
4854	WARNING	Laser: value out of range	Restart the program.
4928	ERROR	Laser: temperature not plausible	Check the chiller.
5010	INFO	Program aborted	Program stopped by operator or by an error.
5017	ERROR	Program: counter changed	Acknowledge and restart.
5055	INFO	Program: counter not plausible	Call service.
5094	WARNING	Program: status changed	Check the central lubrication unit.
5153	ERROR	Program: temperature not plausible	Call service.
5220	INFO	Program: counter not plausible	Restart the program.
5248	INFO	Program: signal out of range	Restart the program.
5286	ERROR	Program: signal changed	Normal operation.
5363	ERROR	Program: value out of range	Restart the program.
5470	INFO	Program: status timeout	Call service.
5570	INFO	Program: signal not plausible	Check the fume extraction unit.
5585	INFO	Program: value missing	Normal operation.
5618	ERROR	Program: temperature changed	Call service.
5636	INFO	Program: counter out of range	Call service.
5656	INFO	Program: value out of range	Call service.
5708	ERROR	Program: status timeout	Call service.

Code	Level	Text	Action
5744	WARNING	Program: temperature not plausible	Restart the program.
5811	INFO	Program: status limit reached	Acknowledge and restart.
5938	INFO	Program: parameter timeout	Restart the program.
6001	INFO	Sheet: status limit reached	Check the pallet changer.
6002	WARNING	Sheet tip-up detected - check part	Tilted part may cause collision. Remove part.
6047	INFO	Sheet: value out of range	Call service.
6191	ERROR	Sheet: value changed	Acknowledge and restart.
6200	ERROR	Sheet: status limit reached	Check the beam guide housing.
6312	INFO	Sheet: counter changed	Acknowledge and restart.
6346	INFO	Sheet: status out of range	Normal operation.
6362	WARNING	Sheet: counter missing	Acknowledge and restart.
6508	INFO	Sheet: counter not plausible	Normal operation.
6670	ERROR	Sheet: status missing	Restart the program.
6816	INFO	Sheet: status timeout	Restart the program.
6817	INFO	Sheet: temperature changed	Check the door interlock.
6900	INFO	Sheet: status missing	Normal operation.
6926	WARNING	Sheet: temperature changed	Call service.
6976	WARNING	Sheet: signal timeout	Normal operation.
6994	INFO	Sheet: signal limit reached	Check the operator console.
7041	INFO	Machine: counter missing	Normal operation.
7062	WARNING	Machine: status not plausible	Check the X axis drive.
7100	INFO	Door opened	Normal operation.
7141	INFO	Machine: value timeout	Acknowledge and restart.
7142	INFO	Machine: temperature out of range	Normal operation.
7157	INFO	Machine: parameter limit reached	Restart the program.
7159	ERROR	Machine: value out of range	Normal operation.
7200	INFO	Pallet change	Normal operation.
7300	INFO	Nozzle centring passed	Normal operation.
7367	WARNING	Machine: status missing	Call service.
7428	INFO	Machine: status out of range	Check the light curtain.
7629	INFO	Machine: value limit reached	Normal operation.
7640	INFO	Machine: status missing	Call service.
7647	ERROR	Machine: parameter missing	Call service.
7694	WARNING	Machine: counter out of range	Acknowledge and restart.
7710	INFO	Machine: counter out of range	Normal operation.
7740	WARNING	Machine: parameter timeout	Call service.
7772	ERROR	Machine: signal missing	Call service.
7801	ERROR	Machine: value missing	Acknowledge and restart.
7809	INFO	Machine: status out of range	Call service.
7861	WARNING	Machine: status out of range	Check the beam guide housing.
8000	INFO	Lubrication interval reminder	Lubricate at end of shift.
8021	WARNING	Maintenance: value missing	Call service.
8095	WARNING	Maintenance: temperature limit reached	Call service.
8123	INFO	Maintenance: status changed	Restart the program.
8159	INFO	Maintenance: value limit reached	Normal operation.
8254	ERROR	Maintenance: counter limit reached	Call service.
8329	INFO	Maintenance: value limit reached	Call service.

Code	Level	Text	Action
8355	INFO	Maintenance: value limit reached	Call service.
8362	ERROR	Maintenance: value timeout	Restart the program.
8466	INFO	Maintenance: value changed	Restart the program.
8500	WARNING	Maintenance: signal timeout	Check the fume extraction unit.
8525	ERROR	Maintenance: signal missing	Normal operation.
8563	WARNING	Maintenance: signal limit reached	Restart the program.
8592	INFO	Maintenance: temperature not plausible	Restart the program.
8612	ERROR	Maintenance: status not plausible	Restart the program.
8718	INFO	Maintenance: counter changed	Call service.
8911	INFO	Maintenance: value changed	Normal operation.
8941	INFO	Maintenance: parameter timeout	Acknowledge and restart.
8994	INFO	Maintenance: temperature changed	Call service.
9024	ERROR	Network: counter changed	Acknowledge and restart.
9027	ERROR	Network: status not plausible	Acknowledge and restart.
9048	WARNING	Network: status limit reached	Restart the program.
9069	INFO	Network: temperature out of range	Call service.
9077	ERROR	Network: parameter not plausible	Restart the program.
9138	INFO	Network: counter missing	Call service.
9218	INFO	Network: parameter missing	Restart the program.
9250	WARNING	Network: signal missing	Normal operation.
9312	ERROR	Network: counter limit reached	Call service.
9363	WARNING	Network: signal limit reached	Acknowledge and restart.
9505	ERROR	Network: counter missing	Call service.
9552	INFO	Network: counter changed	Check the scrap conveyor.
9557	INFO	Network: parameter out of range	Acknowledge and restart.
9607	ERROR	Network: parameter changed	Check the operator console.
9716	INFO	Network: signal timeout	Restart the program.
9787	ERROR	Network: signal not plausible	Acknowledge and restart.
9875	INFO	Network: parameter out of range	Restart the program.
9940	ERROR	Network: signal not plausible	Call service.

10 Troubleshooting cut quality

Inspect the cut edge first. Find the symptom below, then check the causes from top to bottom; the most common cause is listed first. Check the message log (chapter 9) for warnings during the job.

10.1 Dross or burr on the bottom edge

The position of the slat table is shown in the overview drawing in chapter 2. Wear gloves when you touch the slat table; edges can be sharp.

Stainless steel and aluminium (N2).

Rank	Cause	Remedy
1	Cutting gas pressure too low or supply unstable	Check tank, regulator and lines
2	Nozzle worn or damaged (orifice no longer round)	Replace nozzle
3	Focus position wrong	Run focus test
4	Feed rate too high for the thickness	Return to cutting-table feed rate

Mild steel (O2). On mild steel, dross is most often caused by overburning: the feed rate too low (overburning); increase feed rate. The second most likely cause is gas pressure too high; check pressure against cutting table. Less often the material surface rusty or oily; use clean sheet.

10.2 Rough edge or heavy striations

Settings of the slat table are stored in the machine data and should not be changed by the operator. Messages from the slat table are listed in chapter 9 under their code.

Rank	Cause	Remedy
1	Feed rate too high for the thickness	Return to cutting-table feed rate
2	Nozzle worn or damaged (orifice no longer round)	Replace nozzle
3	Protective glass contaminated or damaged (beam power reduced)	Clean or replace protective glass

10.3 Incomplete cut, parts not cut through

Settings of the slat table are stored in the machine data and should not be changed by the operator. The slat table has its own fuse in the control cabinet.

Rank	Cause	Remedy
1	Protective glass contaminated or damaged (beam power reduced)	Clean or replace protective glass
2	Feed rate too high for the thickness	Return to cutting-table feed rate
3	Focus position wrong	Run focus test

NOTE	If message 4001 persists after cleaning the protective glass, contact service.

10.4 Burnt edges or melted corners (thick mild steel)

Dust and spatter shorten the service life of the beam guide housing; keep the area clean. If the beam guide housing does not respond, switch the machine off at the main switch and inform the supervisor.

Check the causes in this order: (1) dirty optics, as in section 10.3 cause 1; (2) feed rate too low (overburning), especially in corners; (3) material too hot (dense nesting); increase part spacing.

10.5 Bevel angle out of tolerance

The position of the slat table is shown in the overview drawing in chapter 2. If the slat table does not respond, switch the machine off at the main switch and inform the supervisor.

Rank	Cause	Remedy
1	Bevel axis / height sensor out of calibration after a collision	Calibrate height sensor and bevel axis
2	Wrong bevel program or tool	Check bevel program; measure first part with protractor

See section 7.4 for the calibration procedure after a collision.

11 Recording a quality problem

11.1 Inspection sheet

Mark the job NOK in the inspection sheet, write the defect type, measure dross height (mm) and roughness Ra (um) if a gauge is available, and take a photo of the cut edge named with the job ID.

11.2 Defect names

Write	Do not write
dross	burr, slag, beard
rough cut	striations, lines
incomplete cut	not through, uncut
burnt edge	burn-out, melted
bevel deviation	angle wrong

12 Service procedures for the maintenance team

This chapter is for the trained maintenance team. Operators do not carry out these procedures.

12.1 Pallet changer

The pallet changer reaches its operating condition a few minutes after the machine is switched on. A label on the pallet changer shows the part number and the year of manufacture. Keep the cover of the pallet changer closed during automatic operation. Changes to the pallet changer must be approved by production engineering. The service technician checks the pallet changer during the yearly inspection.

Spare parts for the pallet changer are listed in Appendix B. Settings of the pallet changer are stored in the machine data and should not be changed by the operator. Wear gloves when you touch the pallet changer; edges can be sharp. The position of the pallet changer is shown in the overview drawing in chapter 2.

The operating hours of the pallet changer are shown on the Diagnostics screen. Only trained personnel may open the pallet changer. The pallet changer has its own fuse in the control cabinet. The pallet changer is monitored by the control; faults are shown on the operator console.

Removal and installation

1. Check that the hoses show no kinks or abrasion.
2. Check that the drive belt has the correct tension.
3. Verify that the hoses show no kinks or abrasion.
4. Inspect that the drive belt has the correct tension.
5. Confirm that the status lamp is green.
6. Check that the fastening screws are in place.
7. Make sure that no foreign objects lie in the travel range.

Adjustment values

Parameter	Nominal	Tolerance
Pallet changer value 1	478 Nm	+/- 10 %
Pallet changer value 2	481 bar	+/- 9 %
Pallet changer value 3	403 s	+/- 5 %
Pallet changer value 4	856 V	+/- 15 %
Pallet changer value 5	334 Nm	+/- 1 %
Pallet changer value 6	50 bar	+/- 6 %

Function test

1. Verify that the drive belt has the correct tension.
2. Check that the hoses show no kinks or abrasion.
3. Make sure that the sensor surface is clean and dry.
4. Make sure that the drive belt has the correct tension.
5. Make sure that the fastening screws are in place.
6. Look at that the cable connections are tight.

Inspection checklist

No.	Check	Every	Done by
1	Inspect that the status lamp is green	month	Maintenance team
2	Make sure that no foreign objects lie in the travel range	year	Service
3	Confirm that the display shows no open messages	500 h	Electrician
4	Confirm that the guard covers are closed and locked	quarter	Maintenance team
5	Inspect that the display shows no open messages	month	Maintenance team
6	Verify that the drive belt has the correct tension	quarter	Service
7	Check that the fastening screws are in place	year	Service

Changes to the pallet changer must be approved by production engineering. If the pallet changer does not respond, switch the machine off at the main switch and inform the supervisor. Wear gloves when you touch the pallet changer; edges can be sharp. The operating hours of the pallet changer are shown on the Diagnostics screen.

12.2 Sheet clamp unit

Record any repair of the sheet clamp unit in the maintenance book. Before working on the sheet clamp unit, switch off the machine and secure the main switch with your own padlock. Changes to the sheet clamp unit must be approved by production engineering. Spare parts for the sheet clamp unit are listed in Appendix B. The sheet clamp unit is designed for continuous operation in a three-shift system.

The sheet clamp unit is monitored by the control; faults are shown on the operator console. The sheet clamp unit was upgraded in 2024; older drawings may show the previous version. The sheet clamp unit can be moved by hand only when the drives are switched off. A label on the sheet clamp unit shows the part number and the year of manufacture.

Record any repair of the sheet clamp unit in the maintenance book. The service technician checks the sheet clamp unit during the yearly inspection. If the sheet clamp unit does not respond, switch the machine off at the main switch and inform the supervisor. After a repair of the sheet clamp unit, run the machine for one test program without a sheet.

Removal and installation

1. Look at that the display shows no open messages.
2. Look at that the status lamp is green.
3. Look at that the display shows no open messages.
4. Verify that the guard covers are closed and locked.
5. Inspect that the display shows no open messages.
6. Make sure that the cable connections are tight.
7. Make sure that the sensor surface is clean and dry.

Adjustment values

Parameter	Nominal	Tolerance
Sheet clamp unit value 1	690 bar	+/- 6 %
Sheet clamp unit value 2	686 mm	+/- 14 %
Sheet clamp unit value 3	551 s	+/- 17 %
Sheet clamp unit value 4	723 bar	+/- 6 %
Sheet clamp unit value 5	287 s	+/- 19 %

Function test

1. Look at that the guard covers are closed and locked.
2. Look at that the filter indicator is in the green range.
3. Inspect that the reference marks line up.
4. Look at that the guard covers are closed and locked.
5. Inspect that the sensor surface is clean and dry.

Inspection checklist

No.	Check	Every	Done by
1	Confirm that the sensor surface is clean and dry	2000 h	Electrician
2	Inspect that the guard covers are closed and locked	year	Maintenance team
3	Look at that the display shows no open messages	2000 h	Electrician
4	Confirm that the fastening screws are in place	2000 h	Electrician
5	Look at that the status lamp is green	month	Electrician
6	Confirm that the fastening screws are in place	month	Maintenance team
7	Inspect that the hoses show no kinks or abrasion	2000 h	Electrician
8	Make sure that the display shows no open messages	500 h	Maintenance team
9	Verify that the filter indicator is in the green range	2000 h	Maintenance team

Spare parts for the sheet clamp unit are listed in Appendix B. The sheet clamp unit is monitored by the control; faults are shown on the operator console. Dust and spatter shorten the service life of the sheet clamp unit; keep the area clean. Keep the cover of the sheet clamp unit closed during automatic operation.

12.3 Door interlock

The door interlock reaches its operating condition a few minutes after the machine is switched on. The door interlock is monitored by the control; faults are shown on the operator console. Keep the cover of the door interlock closed during automatic operation. Changes to the door interlock must be approved by production engineering. If the door interlock makes unusual noise, stop the machine and inform the supervisor.

Record any repair of the door interlock in the maintenance book. After a repair of the door interlock, run the machine for one test program without a sheet. Changes to the door interlock must be approved by production engineering. The door interlock is monitored by the control; faults are shown on the operator console.

The door interlock was upgraded in 2024; older drawings may show the previous version. The operating hours of the door interlock are shown on the Diagnostics screen. The door interlock has its own fuse in the control cabinet. The door interlock is cleaned with a dry cloth; do not use solvents.

Removal and installation

1. Look at that the reference marks line up.
2. Check that the emergency stop is released.
3. Look at that the hoses show no kinks or abrasion.
4. Check that the fastening screws are in place.
5. Look at that the status lamp is green.
6. Verify that the cable connections are tight.
7. Verify that the guard covers are closed and locked.
8. Make sure that the emergency stop is released.
9. Make sure that the status lamp is green.
10. Verify that the hoses show no kinks or abrasion.
11. Check that the guard covers are closed and locked.

Adjustment values

Parameter	Nominal	Tolerance
Door interlock value 1	190 bar	+/- 14 %
Door interlock value 2	534 bar	+/- 18 %
Door interlock value 3	42 V	+/- 9 %
Door interlock value 4	706 mm	+/- 17 %
Door interlock value 5	754 s	+/- 18 %
Door interlock value 6	612 Nm	+/- 16 %

Function test

1. Look at that no foreign objects lie in the travel range.
2. Verify that the status lamp is green.
3. Check that the filter indicator is in the green range.
4. Look at that the reference marks line up.

Inspection checklist

No.	Check	Every	Done by
1	Make sure that the display shows no open messages	year	Electrician
2	Look at that the status lamp is green	2000 h	Service
3	Confirm that the filter indicator is in the green range	month	Electrician
4	Make sure that the filter indicator is in the green range	month	Maintenance team
5	Inspect that the fastening screws are in place	quarter	Electrician
6	Make sure that the fastening screws are in place	2000 h	Maintenance team
7	Check that the cable connections are tight	year	Electrician
8	Check that the sensor surface is clean and dry	year	Service
9	Check that no foreign objects lie in the travel range	year	Service

The door interlock is designed for continuous operation in a three-shift system. The door interlock can be moved by hand only when the drives are switched off. Messages from the door interlock are listed in chapter 9 under their code. The door interlock is cleaned with a dry cloth; do not use solvents.

12.4 Fume extraction unit

Dust and spatter shorten the service life of the fume extraction unit; keep the area clean. The position of the fume extraction unit is shown in the overview drawing in chapter 2. A label on the fume extraction unit shows the part number and the year of manufacture. Only trained personnel may open the fume extraction unit. The fume extraction unit is cleaned with a dry cloth; do not use solvents.

The service technician checks the fume extraction unit during the yearly inspection. The fume extraction unit is monitored by the control; faults are shown on the operator console. The position of the fume extraction unit is shown in the overview drawing in chapter 2. The fume extraction unit is cleaned with a dry cloth; do not use solvents.

The operating hours of the fume extraction unit are shown on the Diagnostics screen. The fume extraction unit was upgraded in 2024; older drawings may show the previous version. Keep the cover of the fume extraction unit closed during automatic operation. A label on the fume extraction unit shows the part number and the year of manufacture.

Removal and installation

1. Check that the fastening screws are in place.
2. Inspect that the emergency stop is released.
3. Verify that the display shows no open messages.
4. Make sure that the emergency stop is released.
5. Verify that the drive belt has the correct tension.
6. Look at that the drive belt has the correct tension.
7. Make sure that the reference marks line up.
8. Look at that the display shows no open messages.

Adjustment values

Parameter	Nominal	Tolerance
Fume extraction unit value 1	837 bar	+/- 19 %
Fume extraction unit value 2	316 V	+/- 18 %
Fume extraction unit value 3	294 mm	+/- 17 %
Fume extraction unit value 4	512 mm	+/- 13 %
Fume extraction unit value 5	68 bar	+/- 18 %
Fume extraction unit value 6	156 bar	+/- 2 %
Fume extraction unit value 7	321 bar	+/- 7 %

Function test

1. Make sure that the emergency stop is released.
2. Verify that the guard covers are closed and locked.
3. Check that no foreign objects lie in the travel range.
4. Look at that the emergency stop is released.
5. Verify that the hoses show no kinks or abrasion.
6. Look at that the status lamp is green.
7. Inspect that the filter indicator is in the green range.

Inspection checklist

No.	Check	Every	Done by
1	Make sure that the emergency stop is released	2000 h	Electrician
2	Inspect that the sensor surface is clean and dry	quarter	Maintenance team
3	Make sure that the filter indicator is in the green range	month	Service
4	Confirm that the cable connections are tight	year	Electrician
5	Look at that the emergency stop is released	year	Service
6	Verify that the drive belt has the correct tension	500 h	Service
7	Confirm that the hoses show no kinks or abrasion	quarter	Service
8	Look at that the cable connections are tight	500 h	Electrician
9	Verify that the sensor surface is clean and dry	year	Service
10	Verify that the status lamp is green	month	Electrician
11	Inspect that the hoses show no kinks or abrasion	year	Maintenance team
12	Look at that the emergency stop is released	2000 h	Electrician

Changes to the fume extraction unit must be approved by production engineering. The fume extraction unit reaches its operating condition a few minutes after the machine is switched on. The fume extraction unit can be moved by hand only when the drives are switched off. The fume extraction unit was upgraded in 2024; older drawings may show the previous version.

12.5 Chiller

If the chiller makes unusual noise, stop the machine and inform the supervisor. The position of the chiller is shown in the overview drawing in chapter 2. Spare parts for the chiller are listed in Appendix B. The operating hours of the chiller are shown on the Diagnostics screen. If the chiller does not respond, switch the machine off at the main switch and inform the supervisor.

The chiller is monitored by the control; faults are shown on the operator console. Record any repair of the chiller in the maintenance book. Keep the cover of the chiller closed during automatic operation. Settings of the chiller are stored in the machine data and should not be changed by the operator.

Settings of the chiller are stored in the machine data and should not be changed by the operator. The service technician checks the chiller during the yearly inspection. After a repair of the chiller, run the machine for one test program without a sheet. The chiller is monitored by the control; faults are shown on the operator console.

Removal and installation

1. Make sure that the display shows no open messages.
2. Confirm that the status lamp is green.
3. Make sure that the display shows no open messages.
4. Verify that the hoses show no kinks or abrasion.
5. Inspect that the fastening screws are in place.
6. Inspect that the sensor surface is clean and dry.
7. Inspect that the emergency stop is released.

Adjustment values

Parameter	Nominal	Tolerance
Chiller value 1	415 bar	+/- 2 %
Chiller value 2	641 Nm	+/- 8 %
Chiller value 3	271 V	+/- 16 %
Chiller value 4	164 s	+/- 10 %

Function test

1. Check that the reference marks line up.
2. Check that the guard covers are closed and locked.
3. Make sure that the reference marks line up.
4. Verify that the guard covers are closed and locked.
5. Check that the sensor surface is clean and dry.

Inspection checklist

No.	Check	Every	Done by
1	Make sure that the sensor surface is clean and dry	year	Maintenance team

No.	Check	Every	Done by
2	Verify that no foreign objects lie in the travel range	month	Service
3	Check that the hoses show no kinks or abrasion	month	Maintenance team
4	Make sure that the guard covers are closed and locked	2000 h	Electrician
5	Inspect that no foreign objects lie in the travel range	year	Service
6	Make sure that no foreign objects lie in the travel range	quarter	Service
7	Confirm that the reference marks line up	year	Maintenance team
8	Look at that the filter indicator is in the green range	year	Service
9	Verify that the emergency stop is released	quarter	Maintenance team
10	Confirm that the status lamp is green	2000 h	Service
11	Verify that the display shows no open messages	quarter	Electrician

The chiller can be moved by hand only when the drives are switched off. Wear gloves when you touch the chiller; edges can be sharp. Keep the cover of the chiller closed during automatic operation. Messages from the chiller are listed in chapter 9 under their code.

12.6 X axis drive

If the X axis drive does not respond, switch the machine off at the main switch and inform the supervisor. After a repair of the X axis drive, run the machine for one test program without a sheet. Record any repair of the X axis drive in the maintenance book. The X axis drive is designed for continuous operation in a three-shift system. Do not use compressed air to clean the X axis drive; it blows dust into the bearings.

The operating hours of the X axis drive are shown on the Diagnostics screen. Record any repair of the X axis drive in the maintenance book. The X axis drive is monitored by the control; faults are shown on the operator console. The service technician checks the X axis drive during the yearly inspection.

The X axis drive is designed for continuous operation in a three-shift system. After a repair of the X axis drive, run the machine for one test program without a sheet. Do not use compressed air to clean the X axis drive; it blows dust into the bearings. The operating hours of the X axis drive are shown on the Diagnostics screen.

Removal and installation

1. Make sure that the sensor surface is clean and dry.
2. Check that the guard covers are closed and locked.
3. Look at that the guard covers are closed and locked.
4. Verify that the emergency stop is released.
5. Make sure that the filter indicator is in the green range.
6. Confirm that the display shows no open messages.
7. Verify that the emergency stop is released.
8. Verify that the drive belt has the correct tension.

Adjustment values

Parameter	Nominal	Tolerance
X axis drive value 1	208 V	+/- 9 %
X axis drive value 2	21 Nm	+/- 8 %
X axis drive value 3	219 Nm	+/- 17 %
X axis drive value 4	90 bar	+/- 1 %

Function test

1. Check that the reference marks line up.
2. Confirm that the reference marks line up.
3. Look at that the reference marks line up.
4. Confirm that the hoses show no kinks or abrasion.

Inspection checklist

No.	Check	Every	Done by
1	Verify that no foreign objects lie in the travel range	month	Maintenance team
2	Inspect that the sensor surface is clean and dry	500 h	Electrician
3	Look at that the guard covers are closed and locked	2000 h	Service
4	Confirm that the sensor surface is clean and dry	year	Service
5	Make sure that the drive belt has the correct tension	2000 h	Electrician
6	Check that the guard covers are closed and locked	quarter	Maintenance team
7	Make sure that the cable connections are tight	2000 h	Electrician
8	Inspect that the guard covers are closed and locked	quarter	Service
9	Confirm that the hoses show no kinks or abrasion	500 h	Electrician

If the X axis drive does not respond, switch the machine off at the main switch and inform the supervisor. Only trained personnel may open the X axis drive. The X axis drive is designed for continuous operation in a three-shift system. Dust and spatter shorten the service life of the X axis drive; keep the area clean.

12.7 Y axis drive

The Y axis drive reaches its operating condition a few minutes after the machine is switched on. The position of the Y axis drive is shown in the overview drawing in chapter 2. A label on the Y axis drive shows the part number and the year of manufacture. The service technician checks the Y axis drive during the yearly inspection. The Y axis drive was upgraded in 2024; older drawings may show the previous version.

The operating hours of the Y axis drive are shown on the Diagnostics screen. Messages from the Y axis drive are listed in chapter 9 under their code. After a repair of the Y axis drive, run the machine for one test program without a sheet. The Y axis drive can be moved by hand only when the drives are switched off.

After a repair of the Y axis drive, run the machine for one test program without a sheet. The service technician checks the Y axis drive during the yearly inspection. Messages from the Y axis drive are listed in chapter 9 under their code. The Y axis drive is designed for continuous operation in a three-shift system.

Removal and installation

1. Confirm that the drive belt has the correct tension.
2. Confirm that the status lamp is green.
3. Inspect that the sensor surface is clean and dry.
4. Verify that the guard covers are closed and locked.
5. Look at that the filter indicator is in the green range.
6. Confirm that the reference marks line up.
7. Check that the guard covers are closed and locked.
8. Check that the drive belt has the correct tension.
9. Look at that the guard covers are closed and locked.

Adjustment values

Parameter	Nominal	Tolerance
Y axis drive value 1	837 Nm	+/- 13 %
Y axis drive value 2	349 s	+/- 9 %
Y axis drive value 3	616 V	+/- 15 %
Y axis drive value 4	800 V	+/- 14 %
Y axis drive value 5	621 s	+/- 14 %
Y axis drive value 6	750 Nm	+/- 17 %

Function test

1. Inspect that the status lamp is green.
2. Make sure that the drive belt has the correct tension.
3. Confirm that no foreign objects lie in the travel range.
4. Check that the display shows no open messages.
5. Look at that the display shows no open messages.
6. Make sure that the filter indicator is in the green range.

Inspection checklist

No.	Check	Every	Done by
1	Make sure that the filter indicator is in the green range	500 h	Maintenance team
2	Inspect that the filter indicator is in the green range	year	Service
3	Inspect that the hoses show no kinks or abrasion	2000 h	Service
4	Inspect that the guard covers are closed and locked	year	Maintenance team
5	Confirm that the guard covers are closed and locked	2000 h	Maintenance team
6	Inspect that the status lamp is green	year	Maintenance team
7	Make sure that the reference marks line up	2000 h	Service
8	Confirm that the status lamp is green	quarter	Service
9	Confirm that the cable connections are tight	quarter	Service
10	Look at that the reference marks line up	500 h	Maintenance team
11	Check that the drive belt has the correct tension	500 h	Electrician
12	Check that no foreign objects lie in the travel range	quarter	Service

The Y axis drive can be moved by hand only when the drives are switched off. The Y axis drive is designed for continuous operation in a three-shift system. The Y axis drive has its own fuse in the control cabinet. Spare parts for the Y axis drive are listed in Appendix B.

12.8 Z axis drive

Record any repair of the Z axis drive in the maintenance book. Before working on the Z axis drive, switch off the machine and secure the main switch with your own padlock. Changes to the Z axis drive must be approved by production engineering. The operating hours of the Z axis drive are shown on the Diagnostics screen. The Z axis drive is designed for continuous operation in a three-shift system.

Record any repair of the Z axis drive in the maintenance book. After a repair of the Z axis drive, run the machine for one test program without a sheet. The Z axis drive reaches its operating condition a few minutes after the machine is switched on. Changes to the Z axis drive must be approved by production engineering.

The Z axis drive reaches its operating condition a few minutes after the machine is switched on. If the Z axis drive does not respond, switch the machine off at the main switch and inform the supervisor. The Z axis drive was upgraded in 2024; older drawings may show the previous version. After a repair of the Z axis drive, run

the machine for one test program without a sheet.

Removal and installation

1. Inspect that the fastening screws are in place.
2. Make sure that no foreign objects lie in the travel range.
3. Make sure that the fastening screws are in place.
4. Confirm that the status lamp is green.
5. Make sure that the guard covers are closed and locked.
6. Verify that the drive belt has the correct tension.
7. Verify that the reference marks line up.

Adjustment values

Parameter	Nominal	Tolerance
Z axis drive value 1	866 mm	+/- 6 %
Z axis drive value 2	491 bar	+/- 18 %
Z axis drive value 3	125 s	+/- 14 %
Z axis drive value 4	541 s	+/- 18 %
Z axis drive value 5	494 Nm	+/- 9 %
Z axis drive value 6	113 mm	+/- 20 %
Z axis drive value 7	866 bar	+/- 10 %
Z axis drive value 8	662 s	+/- 1 %

Function test

1. Verify that the fastening screws are in place.
2. Inspect that the guard covers are closed and locked.
3. Verify that the status lamp is green.
4. Look at that the display shows no open messages.

Inspection checklist

No.	Check	Every	Done by
1	Verify that the hoses show no kinks or abrasion	2000 h	Electrician
2	Verify that the filter indicator is in the green range	500 h	Maintenance team
3	Look at that the filter indicator is in the green range	500 h	Maintenance team
4	Look at that the guard covers are closed and locked	year	Maintenance team
5	Make sure that the fastening screws are in place	month	Maintenance team
6	Check that no foreign objects lie in the travel range	year	Maintenance team
7	Verify that the guard covers are closed and locked	quarter	Service
8	Check that the cable connections are tight	quarter	Service

The Z axis drive is cleaned with a dry cloth; do not use solvents. If the Z axis drive makes unusual noise, stop the machine and inform the supervisor. The Z axis drive is monitored by the control; faults are shown on the operator console. Changes to the Z axis drive must be approved by production engineering.

12.9 Bevel rotation axis

Before working on the bevel rotation axis, switch off the machine and secure the main switch with your own padlock. The bevel rotation axis is monitored by the control; faults are shown on the operator console. Settings of the bevel rotation axis are stored in the machine data and should not be changed by the operator. If the bevel rotation axis does not respond, switch the machine off at the main switch and inform the supervisor. After a repair of the bevel rotation axis, run the machine for one test program without a sheet.

The operating hours of the bevel rotation axis are shown on the Diagnostics screen. Changes to the bevel rotation axis must be approved by production engineering. Only trained personnel may open the bevel rotation axis. Before working on the bevel rotation axis, switch off the machine and secure the main switch with your own padlock.

Keep the cover of the bevel rotation axis closed during automatic operation. Wear gloves when you touch the bevel rotation axis; edges can be sharp. Only trained personnel may open the bevel rotation axis. Record any repair of the bevel rotation axis in the maintenance book.

Removal and installation

1. Inspect that the filter indicator is in the green range.
2. Confirm that the filter indicator is in the green range.
3. Check that the hoses show no kinks or abrasion.
4. Look at that the reference marks line up.
5. Make sure that the reference marks line up.
6. Verify that the display shows no open messages.
7. Make sure that the hoses show no kinks or abrasion.
8. Make sure that the emergency stop is released.
9. Confirm that the drive belt has the correct tension.

Adjustment values

Parameter	Nominal	Tolerance
Bevel rotation axis value 1	182 bar	+/- 13 %
Bevel rotation axis value 2	754 bar	+/- 8 %
Bevel rotation axis value 3	25 s	+/- 14 %
Bevel rotation axis value 4	447 bar	+/- 13 %
Bevel rotation axis value 5	429 Nm	+/- 9 %
Bevel rotation axis value 6	253 V	+/- 16 %
Bevel rotation axis value 7	273 Nm	+/- 19 %

Function test

1. Verify that the emergency stop is released.
2. Inspect that the fastening screws are in place.
3. Check that the emergency stop is released.
4. Look at that the emergency stop is released.
5. Inspect that the reference marks line up.

Inspection checklist

No.	Check	Every	Done by
1	Verify that the sensor surface is clean and dry	quarter	Service
2	Inspect that the hoses show no kinks or abrasion	2000 h	Service
3	Check that the cable connections are tight	quarter	Electrician
4	Look at that no foreign objects lie in the travel range	month	Service
5	Check that the fastening screws are in place	2000 h	Service
6	Verify that the emergency stop is released	500 h	Service
7	Confirm that the emergency stop is released	quarter	Electrician
8	Inspect that the display shows no open messages	quarter	Electrician
9	Look at that the fastening screws are in place	month	Electrician
10	Confirm that the cable connections are tight	500 h	Maintenance team
11	Check that the status lamp is green	2000 h	Electrician
12	Make sure that the sensor surface is clean and dry	year	Electrician

If the bevel rotation axis does not respond, switch the machine off at the main switch and inform the supervisor. The service technician checks the bevel rotation axis during the yearly inspection. The bevel rotation axis can be moved by hand only when the drives are switched off. Keep the cover of the bevel rotation axis closed during automatic operation.

12.10 Slat table

Dust and spatter shorten the service life of the slat table; keep the area clean. Keep the cover of the slat table closed during automatic operation. The slat table has its own fuse in the control cabinet. The slat table is monitored by the control; faults are shown on the operator console. The service technician checks the slat table during the yearly inspection.

Spare parts for the slat table are listed in Appendix B. The slat table has its own fuse in the control cabinet. Changes to the slat table must be approved by production engineering. Settings of the slat table are stored in the machine data and should not be changed by the operator.

Wear gloves when you touch the slat table; edges can be sharp. A label on the slat table shows the part number and the year of manufacture. The operating hours of the slat table are shown on the Diagnostics screen. Only trained personnel may open the slat table.

Removal and installation

1. Check that the hoses show no kinks or abrasion.
2. Inspect that the drive belt has the correct tension.
3. Make sure that the sensor surface is clean and dry.
4. Check that the display shows no open messages.
5. Make sure that the status lamp is green.
6. Make sure that the guard covers are closed and locked.
7. Check that no foreign objects lie in the travel range.
8. Inspect that the display shows no open messages.
9. Look at that the reference marks line up.
10. Verify that the fastening screws are in place.

Adjustment values

Parameter	Nominal	Tolerance
Slat table value 1	415 Nm	+/- 3 %
Slat table value 2	407 bar	+/- 10 %

Parameter	Nominal	Tolerance
Slat table value 3	610 V	+/- 10 %
Slat table value 4	643 V	+/- 11 %
Slat table value 5	734 mm	+/- 3 %
Slat table value 6	433 mm	+/- 20 %
Slat table value 7	790 V	+/- 19 %

Function test

1. Inspect that no foreign objects lie in the travel range.
2. Confirm that the fastening screws are in place.
3. Verify that the drive belt has the correct tension.
4. Make sure that the cable connections are tight.
5. Verify that the filter indicator is in the green range.
6. Confirm that the drive belt has the correct tension.

Inspection checklist

No.	Check	Every	Done by
1	Check that the guard covers are closed and locked	year	Electrician
2	Confirm that the reference marks line up	quarter	Service
3	Confirm that the reference marks line up	quarter	Service
4	Make sure that the sensor surface is clean and dry	2000 h	Maintenance team
5	Make sure that the reference marks line up	500 h	Maintenance team
6	Confirm that the cable connections are tight	quarter	Service
7	Confirm that the filter indicator is in the green range	2000 h	Maintenance team
8	Check that the filter indicator is in the green range	500 h	Maintenance team
9	Look at that the display shows no open messages	month	Maintenance team
10	Inspect that the status lamp is green	2000 h	Electrician
11	Confirm that the sensor surface is clean and dry	500 h	Maintenance team
12	Look at that the cable connections are tight	500 h	Service

Before working on the slat table, switch off the machine and secure the main switch with your own padlock. Messages from the slat table are listed in chapter 9 under their code. Settings of the slat table are stored in the machine data and should not be changed by the operator. Wear gloves when you touch the slat table; edges can be sharp.

12.11 Scrap conveyor

The scrap conveyor reaches its operating condition a few minutes after the machine is switched on. Before working on the scrap conveyor, switch off the machine and secure the main switch with your own padlock. Wear gloves when you touch the scrap conveyor; edges can be sharp. Dust and spatter shorten the service life of the scrap conveyor; keep the area clean. Keep the cover of the scrap conveyor closed during automatic operation.

Spare parts for the scrap conveyor are listed in Appendix B. Dust and spatter shorten the service life of the scrap conveyor; keep the area clean. If the scrap conveyor does not respond, switch the machine off at the main switch and inform the supervisor. The scrap conveyor was upgraded in 2024; older drawings may show the previous version.

The service technician checks the scrap conveyor during the yearly inspection. The scrap conveyor reaches its operating condition a few minutes after the machine is switched on. If the scrap conveyor makes unusual noise, stop the machine and inform the supervisor. Before working on the scrap conveyor, switch off the machine and secure the main switch with your own padlock.

Removal and installation

1. Make sure that the reference marks line up.
2. Make sure that the fastening screws are in place.
3. Inspect that the sensor surface is clean and dry.
4. Verify that the hoses show no kinks or abrasion.
5. Inspect that the guard covers are closed and locked.
6. Look at that the drive belt has the correct tension.
7. Verify that the guard covers are closed and locked.
8. Verify that the drive belt has the correct tension.
9. Check that the display shows no open messages.
10. Make sure that the reference marks line up.
11. Look at that the guard covers are closed and locked.

Adjustment values

Parameter	Nominal	Tolerance
Scrap conveyor value 1	24 V	+/- 17 %
Scrap conveyor value 2	612 s	+/- 9 %
Scrap conveyor value 3	131 V	+/- 9 %
Scrap conveyor value 4	885 Nm	+/- 10 %

Function test

1. Check that the sensor surface is clean and dry.
2. Look at that no foreign objects lie in the travel range.
3. Inspect that the sensor surface is clean and dry.
4. Look at that the drive belt has the correct tension.

Inspection checklist

No.	Check	Every	Done by
1	Make sure that the fastening screws are in place	year	Electrician
2	Make sure that the status lamp is green	year	Service
3	Check that the hoses show no kinks or abrasion	2000 h	Maintenance team
4	Confirm that the reference marks line up	year	Electrician
5	Make sure that the fastening screws are in place	quarter	Electrician
6	Look at that the status lamp is green	quarter	Electrician
7	Verify that the drive belt has the correct tension	month	Service
8	Look at that the drive belt has the correct tension	500 h	Service
9	Inspect that the sensor surface is clean and dry	year	Service

Settings of the scrap conveyor are stored in the machine data and should not be changed by the operator. After a repair of the scrap conveyor, run the machine for one test program without a sheet. Only trained personnel may open the scrap conveyor. The position of the scrap conveyor is shown in the overview drawing in chapter 2.

12.12 Operator console

Record any repair of the operator console in the maintenance book. If the operator console makes unusual noise, stop the machine and inform the supervisor. The operator console is monitored by the control; faults are shown on the operator console. Settings of the operator console are stored in the machine data and should not be changed by the operator. Only trained personnel may open the operator console.

The position of the operator console is shown in the overview drawing in chapter 2. The operator console is monitored by the control; faults are shown on the operator console. Before working on the operator console, switch off the machine and secure the main switch with your own padlock. The operator console was upgraded in 2024; older drawings may show the previous version.

If the operator console does not respond, switch the machine off at the main switch and inform the supervisor. The operator console is designed for continuous operation in a three-shift system. If the operator console makes unusual noise, stop the machine and inform the supervisor. The operating hours of the operator console are shown on the Diagnostics screen.

Removal and installation

1. Make sure that the filter indicator is in the green range.
2. Check that the cable connections are tight.
3. Confirm that the drive belt has the correct tension.
4. Check that the hoses show no kinks or abrasion.
5. Confirm that the display shows no open messages.
6. Inspect that the reference marks line up.
7. Inspect that the status lamp is green.
8. Verify that the sensor surface is clean and dry.

Adjustment values

Parameter	Nominal	Tolerance
Operator console value 1	428 s	+/- 12 %
Operator console value 2	483 bar	+/- 17 %
Operator console value 3	573 s	+/- 5 %
Operator console value 4	831 bar	+/- 13 %
Operator console value 5	187 Nm	+/- 5 %
Operator console value 6	745 s	+/- 16 %
Operator console value 7	152 bar	+/- 1 %
Operator console value 8	4 s	+/- 3 %

Function test

1. Inspect that the guard covers are closed and locked.
2. Look at that the drive belt has the correct tension.
3. Verify that the cable connections are tight.
4. Verify that the filter indicator is in the green range.

5. Inspect that the fastening screws are in place.

6. Inspect that the status lamp is green.

7. Make sure that the display shows no open messages.

Inspection checklist

No.	Check	Every	Done by
1	Check that the reference marks line up	year	Service
2	Verify that the cable connections are tight	2000 h	Service
3	Inspect that the guard covers are closed and locked	quarter	Service
4	Check that the fastening screws are in place	year	Electrician
5	Inspect that no foreign objects lie in the travel range	year	Service
6	Inspect that the reference marks line up	500 h	Maintenance team
7	Confirm that the emergency stop is released	quarter	Maintenance team
8	Verify that the filter indicator is in the green range	500 h	Electrician

The operator console is cleaned with a dry cloth; do not use solvents. After a repair of the operator console, run the machine for one test program without a sheet. Record any repair of the operator console in the maintenance book. The operator console has its own fuse in the control cabinet.

12.13 Beam guide housing

A label on the beam guide housing shows the part number and the year of manufacture. The beam guide housing can be moved by hand only when the drives are switched off. The beam guide housing has its own fuse in the control cabinet. The operating hours of the beam guide housing are shown on the Diagnostics screen. The position of the beam guide housing is shown in the overview drawing in chapter 2.

The beam guide housing is designed for continuous operation in a three-shift system. Record any repair of the beam guide housing in the maintenance book. If the beam guide housing makes unusual noise, stop the machine and inform the supervisor. Keep the cover of the beam guide housing closed during automatic operation.

Keep the cover of the beam guide housing closed during automatic operation. Dust and spatter shorten the service life of the beam guide housing; keep the area clean. The position of the beam guide housing is shown in the overview drawing in chapter 2. A label on the beam guide housing shows the part number and the year of manufacture.

Removal and installation

1. Inspect that the status lamp is green.

2. Make sure that the fastening screws are in place.

3. Verify that the display shows no open messages.

4. Make sure that the display shows no open messages.

5. Confirm that the cable connections are tight.

6. Confirm that the hoses show no kinks or abrasion.

7. Look at that the reference marks line up.

8. Check that the guard covers are closed and locked.

9. Look at that the reference marks line up.

10. Check that the display shows no open messages.

11. Make sure that no foreign objects lie in the travel range.

Adjustment values

Parameter	Nominal	Tolerance
Beam guide housing value 1	259 bar	+/- 20 %
Beam guide housing value 2	367 bar	+/- 17 %
Beam guide housing value 3	709 Nm	+/- 7 %
Beam guide housing value 4	342 mm	+/- 11 %
Beam guide housing value 5	229 Nm	+/- 16 %
Beam guide housing value 6	888 mm	+/- 19 %

Function test

1. Check that no foreign objects lie in the travel range.
2. Check that the fastening screws are in place.
3. Confirm that the sensor surface is clean and dry.
4. Check that the cable connections are tight.
5. Verify that the fastening screws are in place.

Inspection checklist

No.	Check	Every	Done by
1	Confirm that the hoses show no kinks or abrasion	2000 h	Service
2	Make sure that the hoses show no kinks or abrasion	500 h	Electrician
3	Confirm that the guard covers are closed and locked	year	Service
4	Confirm that no foreign objects lie in the travel range	year	Electrician
5	Verify that the emergency stop is released	month	Service
6	Check that the reference marks line up	quarter	Electrician
7	Inspect that the display shows no open messages	year	Service
8	Verify that the emergency stop is released	month	Electrician
9	Inspect that the filter indicator is in the green range	year	Service
10	Confirm that the hoses show no kinks or abrasion	quarter	Maintenance team
11	Verify that the sensor surface is clean and dry	2000 h	Maintenance team

Changes to the beam guide housing must be approved by production engineering. The beam guide housing is monitored by the control; faults are shown on the operator console. Only trained personnel may open the beam guide housing. The beam guide housing can be moved by hand only when the drives are switched off.

12.14 Compressed air unit

Do not use compressed air to clean the compressed air unit; it blows dust into the bearings. Dust and spatter shorten the service life of the compressed air unit; keep the area clean. Spare parts for the compressed air unit are listed in Appendix B. Settings of the compressed air unit are stored in the machine data and should not be changed by the operator. If the compressed air unit makes unusual noise, stop the machine and inform the supervisor.

Wear gloves when you touch the compressed air unit; edges can be sharp. Only trained personnel may open the compressed air unit. Dust and spatter shorten the service life of the compressed air unit; keep the area clean. The compressed air unit is cleaned with a dry cloth; do not use solvents.

The compressed air unit was upgraded in 2024; older drawings may show the previous version. The compressed air unit is cleaned with a dry cloth; do not use solvents. A label on the compressed air unit shows the part number and the year of manufacture. Messages from the compressed air unit are listed in chapter 9 under their code.

Removal and installation

1. Confirm that no foreign objects lie in the travel range.
2. Inspect that the cable connections are tight.
3. Inspect that the sensor surface is clean and dry.
4. Check that the filter indicator is in the green range.
5. Look at that the reference marks line up.
6. Inspect that the guard covers are closed and locked.
7. Inspect that the filter indicator is in the green range.
8. Check that the cable connections are tight.
9. Look at that no foreign objects lie in the travel range.

Adjustment values

Parameter	Nominal	Tolerance
Compressed air unit value 1	788 V	+/- 8 %
Compressed air unit value 2	479 mm	+/- 14 %
Compressed air unit value 3	259 bar	+/- 10 %
Compressed air unit value 4	211 mm	+/- 19 %
Compressed air unit value 5	766 V	+/- 7 %
Compressed air unit value 6	842 mm	+/- 14 %

Function test

1. Check that the guard covers are closed and locked.
2. Verify that the drive belt has the correct tension.
3. Verify that the status lamp is green.
4. Inspect that the sensor surface is clean and dry.

Inspection checklist

No.	Check	Every	Done by
1	Make sure that the sensor surface is clean and dry	500 h	Electrician
2	Inspect that the cable connections are tight	2000 h	Electrician
3	Inspect that no foreign objects lie in the travel range	500 h	Electrician
4	Make sure that the drive belt has the correct tension	500 h	Electrician
5	Look at that the filter indicator is in the green range	month	Maintenance team
6	Look at that the display shows no open messages	month	Service
7	Check that the filter indicator is in the green range	2000 h	Maintenance team

The compressed air unit was upgraded in 2024; older drawings may show the previous version. Changes to the compressed air unit must be approved by production engineering. Dust and spatter shorten the service life of the compressed air unit; keep the area clean. The service technician checks the compressed air unit during the yearly inspection.

12.15 Control cabinet

The control cabinet was upgraded in 2024; older drawings may show the previous version. The control cabinet can be moved by hand only when the drives are switched off. The control cabinet has its own fuse in the control cabinet. After a repair of the control cabinet, run the machine for one test program without a sheet. Do not use compressed air to clean the control cabinet; it blows dust into the bearings.

The control cabinet is monitored by the control; faults are shown on the operator console. Only trained personnel may open the control cabinet. A label on the control cabinet shows the part number and the year of manufacture. Record any repair of the control cabinet in the maintenance book.

Before working on the control cabinet, switch off the machine and secure the main switch with your own padlock. The control cabinet is cleaned with a dry cloth; do not use solvents. Dust and spatter shorten the service life of the control cabinet; keep the area clean. The control cabinet can be moved by hand only when the drives are switched off.

Removal and installation

1. Inspect that the guard covers are closed and locked.
2. Verify that no foreign objects lie in the travel range.
3. Confirm that the drive belt has the correct tension.
4. Inspect that the guard covers are closed and locked.
5. Verify that the hoses show no kinks or abrasion.
6. Make sure that no foreign objects lie in the travel range.
7. Look at that the guard covers are closed and locked.
8. Confirm that the reference marks line up.
9. Inspect that the guard covers are closed and locked.
10. Check that the filter indicator is in the green range.
11. Look at that the fastening screws are in place.

Adjustment values

Parameter	Nominal	Tolerance
Control cabinet value 1	732 s	+/- 13 %
Control cabinet value 2	526 mm	+/- 20 %
Control cabinet value 3	352 V	+/- 8 %
Control cabinet value 4	503 mm	+/- 16 %
Control cabinet value 5	92 Nm	+/- 15 %
Control cabinet value 6	892 bar	+/- 8 %
Control cabinet value 7	494 Nm	+/- 17 %
Control cabinet value 8	78 bar	+/- 9 %

Function test

1. Verify that the cable connections are tight.
2. Inspect that the reference marks line up.
3. Verify that the reference marks line up.
4. Inspect that the emergency stop is released.
5. Check that the drive belt has the correct tension.
6. Inspect that the filter indicator is in the green range.

Inspection checklist

No.	Check	Every	Done by
1	Check that the reference marks line up	month	Service
2	Inspect that the hoses show no kinks or abrasion	quarter	Service
3	Make sure that the cable connections are tight	quarter	Maintenance team
4	Confirm that the display shows no open messages	quarter	Electrician
5	Inspect that no foreign objects lie in the travel range	month	Maintenance team
6	Confirm that the status lamp is green	year	Electrician
7	Check that the drive belt has the correct tension	month	Electrician

The control cabinet reaches its operating condition a few minutes after the machine is switched on. Spare parts for the control cabinet are listed in Appendix B. The control cabinet is designed for continuous operation in a three-shift system. Messages from the control cabinet are listed in chapter 9 under their code.

12.16 Light curtain

Record any repair of the light curtain in the maintenance book. The operating hours of the light curtain are shown on the Diagnostics screen. The light curtain was upgraded in 2024; older drawings may show the previous version. A label on the light curtain shows the part number and the year of manufacture. Only trained personnel may open the light curtain.

The light curtain was upgraded in 2024; older drawings may show the previous version. Keep the cover of the light curtain closed during automatic operation. Settings of the light curtain are stored in the machine data and should not be changed by the operator. Do not use compressed air to clean the light curtain; it blows dust into the bearings.

The light curtain reaches its operating condition a few minutes after the machine is switched on. The service technician checks the light curtain during the yearly inspection. Changes to the light curtain must be approved by production engineering. The light curtain has its own fuse in the control cabinet.

Removal and installation

1. Check that the drive belt has the correct tension.
2. Inspect that the display shows no open messages.
3. Confirm that the emergency stop is released.
4. Verify that the fastening screws are in place.
5. Look at that the drive belt has the correct tension.
6. Check that the filter indicator is in the green range.
7. Make sure that the cable connections are tight.
8. Inspect that the hoses show no kinks or abrasion.
9. Look at that the sensor surface is clean and dry.
10. Confirm that the emergency stop is released.

Adjustment values

Parameter	Nominal	Tolerance
Light curtain value 1	222 mm	+/- 13 %
Light curtain value 2	367 Nm	+/- 15 %
Light curtain value 3	759 V	+/- 11 %
Light curtain value 4	552 bar	+/- 16 %

Parameter	Nominal	Tolerance
Light curtain value 5	663 mm	+/- 2 %
Light curtain value 6	432 Nm	+/- 13 %
Light curtain value 7	748 Nm	+/- 6 %

Function test

1. Inspect that the sensor surface is clean and dry.
2. Confirm that the cable connections are tight.
3. Confirm that the filter indicator is in the green range.
4. Check that the emergency stop is released.
5. Check that the display shows no open messages.

Inspection checklist

No.	Check	Every	Done by
1	Make sure that the emergency stop is released	500 h	Maintenance team
2	Check that no foreign objects lie in the travel range	year	Service
3	Look at that the display shows no open messages	month	Service
4	Look at that the reference marks line up	year	Electrician
5	Look at that the cable connections are tight	year	Electrician
6	Check that no foreign objects lie in the travel range	year	Service
7	Confirm that the display shows no open messages	year	Maintenance team
8	Look at that the sensor surface is clean and dry	2000 h	Service
9	Check that the hoses show no kinks or abrasion	month	Electrician
10	Verify that the reference marks line up	year	Service
11	Inspect that the display shows no open messages	month	Service

The light curtain reaches its operating condition a few minutes after the machine is switched on. Only trained personnel may open the light curtain. Keep the cover of the light curtain closed during automatic operation. Changes to the light curtain must be approved by production engineering.

12.17 Part removal flap

The position of the part removal flap is shown in the overview drawing in chapter 2. A label on the part removal flap shows the part number and the year of manufacture. The part removal flap is designed for continuous operation in a three-shift system. The part removal flap is monitored by the control; faults are shown on the operator console. Dust and spatter shorten the service life of the part removal flap; keep the area clean.

The part removal flap is monitored by the control; faults are shown on the operator console. The position of the part removal flap is shown in the overview drawing in chapter 2. Wear gloves when you touch the part removal flap; edges can be sharp. If the part removal flap makes unusual noise, stop the machine and inform the supervisor.

Wear gloves when you touch the part removal flap; edges can be sharp. The part removal flap is monitored by the control; faults are shown on the operator console. If the part removal flap makes unusual noise, stop the machine and inform the supervisor. Do not use compressed air to clean the part removal flap; it blows dust into the bearings.

Removal and installation

1. Look at that the cable connections are tight.

2. Look at that the fastening screws are in place.

3. Inspect that the drive belt has the correct tension.

4. Inspect that the display shows no open messages.

5. Verify that the guard covers are closed and locked.

6. Inspect that the emergency stop is released.

7. Verify that no foreign objects lie in the travel range.

Adjustment values

Parameter	Nominal	Tolerance
Part removal flap value 1	226 s	+/- 7 %
Part removal flap value 2	589 V	+/- 11 %
Part removal flap value 3	466 Nm	+/- 10 %
Part removal flap value 4	176 Nm	+/- 18 %
Part removal flap value 5	692 s	+/- 12 %
Part removal flap value 6	589 V	+/- 1 %
Part removal flap value 7	465 V	+/- 16 %

Function test

1. Inspect that the filter indicator is in the green range.

2. Make sure that no foreign objects lie in the travel range.

3. Inspect that no foreign objects lie in the travel range.

4. Look at that the emergency stop is released.

5. Confirm that the hoses show no kinks or abrasion.

6. Check that the display shows no open messages.

Inspection checklist

No.	Check	Every	Done by
1	Inspect that the guard covers are closed and locked	500 h	Maintenance team
2	Confirm that the display shows no open messages	500 h	Service
3	Check that the fastening screws are in place	500 h	Service
4	Confirm that the display shows no open messages	month	Service
5	Inspect that no foreign objects lie in the travel range	month	Service
6	Confirm that the guard covers are closed and locked	2000 h	Electrician
7	Confirm that no foreign objects lie in the travel range	month	Service

The part removal flap is cleaned with a dry cloth; do not use solvents. Keep the cover of the part removal flap closed during automatic operation. A label on the part removal flap shows the part number and the year of manufacture. Changes to the part removal flap must be approved by production engineering.

12.18 Cable carrier

Keep the cover of the cable carrier closed during automatic operation. A label on the cable carrier shows the part number and the year of manufacture. If the cable carrier does not respond, switch the machine off at the main switch and inform the supervisor. After a repair of the cable carrier, run the machine for one test program without a sheet. The position of the cable carrier is shown in the overview drawing in chapter 2.

Spare parts for the cable carrier are listed in Appendix B. Record any repair of the cable carrier in the maintenance book. The service technician checks the cable carrier during the yearly inspection. The operating hours of the cable carrier are shown on the Diagnostics screen.

The cable carrier was upgraded in 2024; older drawings may show the previous version. Spare parts for the cable carrier are listed in Appendix B. Wear gloves when you touch the cable carrier; edges can be sharp. The service technician checks the cable carrier during the yearly inspection.

Removal and installation

1. Inspect that the guard covers are closed and locked.
2. Make sure that the display shows no open messages.
3. Inspect that the display shows no open messages.
4. Verify that the drive belt has the correct tension.
5. Verify that the hoses show no kinks or abrasion.
6. Verify that the emergency stop is released.
7. Confirm that the drive belt has the correct tension.
8. Verify that the display shows no open messages.
9. Make sure that the guard covers are closed and locked.
10. Check that the status lamp is green.
11. Verify that the guard covers are closed and locked.

Adjustment values

Parameter	Nominal	Tolerance
Cable carrier value 1	279 Nm	+/- 4 %
Cable carrier value 2	232 V	+/- 17 %
Cable carrier value 3	354 Nm	+/- 18 %
Cable carrier value 4	277 s	+/- 13 %
Cable carrier value 5	51 mm	+/- 16 %
Cable carrier value 6	19 s	+/- 3 %
Cable carrier value 7	208 V	+/- 12 %
Cable carrier value 8	67 V	+/- 15 %

Function test

1. Make sure that the display shows no open messages.
2. Look at that the hoses show no kinks or abrasion.
3. Inspect that the fastening screws are in place.
4. Make sure that the drive belt has the correct tension.
5. Check that the status lamp is green.
6. Look at that the drive belt has the correct tension.
7. Make sure that the hoses show no kinks or abrasion.

Inspection checklist

No.	Check	Every	Done by
1	Make sure that the sensor surface is clean and dry	quarter	Maintenance team
2	Verify that the sensor surface is clean and dry	month	Service
3	Verify that the sensor surface is clean and dry	2000 h	Maintenance team
4	Check that the reference marks line up	month	Electrician
5	Inspect that the cable connections are tight	month	Service
6	Make sure that the drive belt has the correct tension	2000 h	Maintenance team
7	Confirm that the emergency stop is released	2000 h	Maintenance team
8	Look at that the cable connections are tight	month	Maintenance team
9	Check that the guard covers are closed and locked	quarter	Maintenance team
10	Verify that the filter indicator is in the green range	2000 h	Maintenance team

Dust and spatter shorten the service life of the cable carrier; keep the area clean. Spare parts for the cable carrier are listed in Appendix B. The operating hours of the cable carrier are shown on the Diagnostics screen. Before working on the cable carrier, switch off the machine and secure the main switch with your own padlock.

12.19 Central lubrication unit

Changes to the central lubrication unit must be approved by production engineering. Before working on the central lubrication unit, switch off the machine and secure the main switch with your own padlock. The central lubrication unit has its own fuse in the control cabinet. The central lubrication unit is cleaned with a dry cloth; do not use solvents. Wear gloves when you touch the central lubrication unit; edges can be sharp.

Wear gloves when you touch the central lubrication unit; edges can be sharp. Record any repair of the central lubrication unit in the maintenance book. Only trained personnel may open the central lubrication unit. Do not use compressed air to clean the central lubrication unit; it blows dust into the bearings.

Spare parts for the central lubrication unit are listed in Appendix B. If the central lubrication unit makes unusual noise, stop the machine and inform the supervisor. Do not use compressed air to clean the central lubrication unit; it blows dust into the bearings. If the central lubrication unit does not respond, switch the machine off at the main switch and inform the supervisor.

Removal and installation

1. Look at that the emergency stop is released.
2. Confirm that the emergency stop is released.
3. Verify that no foreign objects lie in the travel range.
4. Verify that the emergency stop is released.
5. Look at that the cable connections are tight.
6. Make sure that the emergency stop is released.
7. Confirm that the fastening screws are in place.
8. Make sure that the display shows no open messages.
9. Make sure that the emergency stop is released.
10. Look at that no foreign objects lie in the travel range.

Adjustment values

Parameter	Nominal	Tolerance
Central lubrication unit value 1	193 s	+/- 10 %
Central lubrication unit value 2	109 bar	+/- 9 %
Central lubrication unit value 3	557 mm	+/- 1 %

Parameter	Nominal	Tolerance
Central lubrication unit value 4	833 s	+/- 4 %

Function test

1. Verify that the drive belt has the correct tension.
2. Verify that the display shows no open messages.
3. Confirm that the reference marks line up.
4. Verify that the sensor surface is clean and dry.
5. Make sure that the guard covers are closed and locked.
6. Check that the fastening screws are in place.

Inspection checklist

No.	Check	Every	Done by
1	Verify that the hoses show no kinks or abrasion	500 h	Service
2	Verify that the status lamp is green	2000 h	Maintenance team
3	Inspect that no foreign objects lie in the travel range	500 h	Maintenance team
4	Inspect that the hoses show no kinks or abrasion	month	Service
5	Verify that the guard covers are closed and locked	2000 h	Service
6	Check that the filter indicator is in the green range	quarter	Electrician
7	Confirm that the guard covers are closed and locked	2000 h	Service

Before working on the central lubrication unit, switch off the machine and secure the main switch with your own padlock. The operating hours of the central lubrication unit are shown on the Diagnostics screen. Messages from the central lubrication unit are listed in chapter 9 under their code. Keep the cover of the central lubrication unit closed during automatic operation.

12.20 Network interface

The service technician checks the network interface during the yearly inspection. Settings of the network interface are stored in the machine data and should not be changed by the operator. The network interface is monitored by the control; faults are shown on the operator console. A label on the network interface shows the part number and the year of manufacture. Before working on the network interface, switch off the machine and secure the main switch with your own padlock.

A label on the network interface shows the part number and the year of manufacture. Changes to the network interface must be approved by production engineering. Record any repair of the network interface in the maintenance book. Before working on the network interface, switch off the machine and secure the main switch with your own padlock.

Spare parts for the network interface are listed in Appendix B. The network interface was upgraded in 2024; older drawings may show the previous version. Do not use compressed air to clean the network interface; it blows dust into the bearings. The operating hours of the network interface are shown on the Diagnostics screen.

Removal and installation

1. Confirm that the drive belt has the correct tension.
2. Verify that the guard covers are closed and locked.
3. Look at that the status lamp is green.
4. Make sure that the display shows no open messages.
5. Confirm that no foreign objects lie in the travel range.

6. Make sure that the emergency stop is released.

7. Check that the hoses show no kinks or abrasion.

8. Inspect that the filter indicator is in the green range.

9. Confirm that the drive belt has the correct tension.

10. Look at that the display shows no open messages.

Adjustment values

Parameter	Nominal	Tolerance
Network interface value 1	59 Nm	+/- 5 %
Network interface value 2	379 s	+/- 8 %
Network interface value 3	643 s	+/- 10 %
Network interface value 4	487 V	+/- 17 %
Network interface value 5	692 V	+/- 1 %
Network interface value 6	111 Nm	+/- 20 %

Function test

1. Look at that the drive belt has the correct tension.

2. Inspect that the guard covers are closed and locked.

3. Inspect that the status lamp is green.

4. Verify that the guard covers are closed and locked.

5. Verify that the guard covers are closed and locked.

Inspection checklist

No.	Check	Every	Done by
1	Verify that the filter indicator is in the green range	month	Electrician
2	Make sure that the filter indicator is in the green range	500 h	Service
3	Check that the guard covers are closed and locked	quarter	Maintenance team
4	Check that the filter indicator is in the green range	2000 h	Service
5	Confirm that the emergency stop is released	2000 h	Electrician
6	Look at that the status lamp is green	quarter	Service
7	Make sure that the filter indicator is in the green range	2000 h	Service
8	Confirm that the status lamp is green	month	Maintenance team

Settings of the network interface are stored in the machine data and should not be changed by the operator. Record any repair of the network interface in the maintenance book. A label on the network interface shows the part number and the year of manufacture. Wear gloves when you touch the network interface; edges can be sharp.

A Glossary

Term	Meaning
Dross	Molten metal stuck to the bottom edge.
Striations	Fine lines on the cut edge.
Kerf	The gap cut by the beam.
Bevel	A cut at an angle to the sheet surface.
Protective glass	Window that protects the focusing lens from spatter.
Nozzle	Copper tip for gas and beam.
Focus position	Height of the beam focus relative to the sheet surface.
NOK	Not OK, failed inspection.
Pallet changer	Do not use compressed air to clean the pallet changer; it blows dust into the bearings.
Sheet clamp unit	Do not use compressed air to clean the sheet clamp unit; it blows dust into the bearings.
Door interlock	The position of the door interlock is shown in the overview drawing in chapter 2.
Fume extraction unit	The service technician checks the fume extraction unit during the yearly inspection.
Chiller	The chiller has its own fuse in the control cabinet.
X axis drive	Only trained personnel may open the X axis drive.
Y axis drive	Spare parts for the Y axis drive are listed in Appendix B.
Z axis drive	Only trained personnel may open the Z axis drive.
Bevel rotation axis	The position of the bevel rotation axis is shown in the overview drawing in chapter 2.
Slat table	A label on the slat table shows the part number and the year of manufacture.
Scrap conveyor	If the scrap conveyor does not respond, switch the machine off at the main switch and inform the supervisor.
Operator console	Changes to the operator console must be approved by production engineering.
Beam guide housing	Wear gloves when you touch the beam guide housing; edges can be sharp.
Compressed air unit	If the compressed air unit makes unusual noise, stop the machine and inform the supervisor.
Control cabinet	Messages from the control cabinet are listed in chapter 9 under their code.
Light curtain	The light curtain has its own fuse in the control cabinet.
Part removal flap	Keep the cover of the part removal flap closed during automatic operation.
Cable carrier	The operating hours of the cable carrier are shown on the Diagnostics screen.
Central lubrication unit	The central lubrication unit was upgraded in 2024; older drawings may show the previous version.
Network interface	The network interface was upgraded in 2024; older drawings may show the previous version.

B Spare parts

Part	Number	Keep in stock
Nozzle single 1.2 mm	N-120	-
Nozzle single 1.4 mm	N-140	-
Nozzle single 1.7 mm	N-170	-
Nozzle double 2.0 mm	N-200	-
Nozzle double 2.3 mm	N-230	yes
Nozzle double 2.7 mm	N-270	yes
Nozzle double 3.0 mm	N-300	-
Protective glass	PG-01	yes
Ceramic ring	CR-01	-
Pallet changer part 1	SP-2059	-
Pallet changer part 2	SP-2860	-
Pallet changer part 3	SP-8919	-
Sheet clamp unit part 1	SP-2471	-
Sheet clamp unit part 2	SP-7590	-
Sheet clamp unit part 3	SP-1829	-
Door interlock part 1	SP-4523	-
Door interlock part 2	SP-1504	-
Door interlock part 3	SP-4348	-
Fume extraction unit part 1	SP-5862	-
Fume extraction unit part 2	SP-6138	-
Fume extraction unit part 3	SP-6251	-
Chiller part 1	SP-6528	-
Chiller part 2	SP-7901	-
Chiller part 3	SP-6143	-
X axis drive part 1	SP-5632	-
X axis drive part 2	SP-4037	-
X axis drive part 3	SP-6126	-
Y axis drive part 1	SP-2775	-
Y axis drive part 2	SP-7149	-
Y axis drive part 3	SP-8522	-
Z axis drive part 1	SP-9934	-
Z axis drive part 2	SP-4909	-
Z axis drive part 3	SP-5839	-
Bevel rotation axis part 1	SP-5099	-
Bevel rotation axis part 2	SP-2377	-
Bevel rotation axis part 3	SP-6394	-
Slat table part 1	SP-8376	-
Slat table part 2	SP-3176	-
Slat table part 3	SP-7098	-
Scrap conveyor part 1	SP-7989	-
Scrap conveyor part 2	SP-5545	-
Scrap conveyor part 3	SP-9533	-
Operator console part 1	SP-4344	-
Operator console part 2	SP-8490	-
Operator console part 3	SP-6563	-

Part	Number	Keep in stock
Beam guide housing part 1	SP-2079	-
Beam guide housing part 2	SP-8340	-
Beam guide housing part 3	SP-8052	-
Compressed air unit part 1	SP-1244	-
Compressed air unit part 2	SP-8660	-
Compressed air unit part 3	SP-7238	-
Control cabinet part 1	SP-3072	-
Control cabinet part 2	SP-8375	-
Control cabinet part 3	SP-8037	-
Light curtain part 1	SP-9458	-
Light curtain part 2	SP-8983	-
Light curtain part 3	SP-4755	-
Part removal flap part 1	SP-1892	-
Part removal flap part 2	SP-1044	-
Part removal flap part 3	SP-8583	-
Cable carrier part 1	SP-5220	-
Cable carrier part 2	SP-8492	-
Cable carrier part 3	SP-5951	-
Central lubrication unit part 1	SP-7089	-
Central lubrication unit part 2	SP-4813	-
Central lubrication unit part 3	SP-1560	-
Network interface part 1	SP-4669	-
Network interface part 2	SP-5382	-
Network interface part 3	SP-1331	-

C Cell 1 machine (CO2 laser) - legacy values

WARNING	This appendix is for the older Cell 1 CO2 machine. It does NOT apply to Laser Cell 2. Kept for the shared maintenance team.

C.1 About Cell 1

Cell 1 uses a 4 kW CO2 laser with mirrors in the beam path. Its message codes partly overlap with Cell 2.

C.2 Troubleshooting (Cell 1)

Symptom	Possible causes (most likely first)	Remedy
Dross / burr on the bottom edge (N2)	1. Focus position wrong 2. Cutting gas pressure too low 3. Nozzle worn	Run focus test. Check gas. Replace nozzle.
Incomplete cut	1. Mirror contaminated 2. Feed rate too high	Clean mirrors. Reduce feed.
Rough edge	1. Feed rate too high 2. Beam mode degraded	Reduce feed. Call service.

C.3 Maintenance (Cell 1)

Item	Interval
Nozzle	Replace every 120 operating hours
Lens	Clean weekly
Mirrors	Check monthly
Feed rate	Up to 20 % above table allowed for CO2 thin sheet

C.4 Messages (Cell 1)

Code	Level	Text
2105	WARNING	Lens contamination - clean lens
4001	WARNING	Beam mode degraded
1201	WARNING	Cutting gas pressure below setpoint